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(19) **United States**(12) **Patent Application Publication**
Wilks(10) **Pub. No.: US 2025/0283312 A1**(43) **Pub. Date: Sep. 11, 2025**(54) **WASHABLE FAUCET LEVER WITH
CRANK-ACTIVATED DETERGENT
DISPENSER**(52) **U.S. Cl.**CPC *E03C 1/046* (2013.01); *A47K 5/1204*
(2013.01); *A47K 2005/1218* (2013.01)(71) Applicant: **Freeman Wilks**, Sunrise, FL (US)(72) Inventor: **Freeman Wilks**, Sunrise, FL (US)

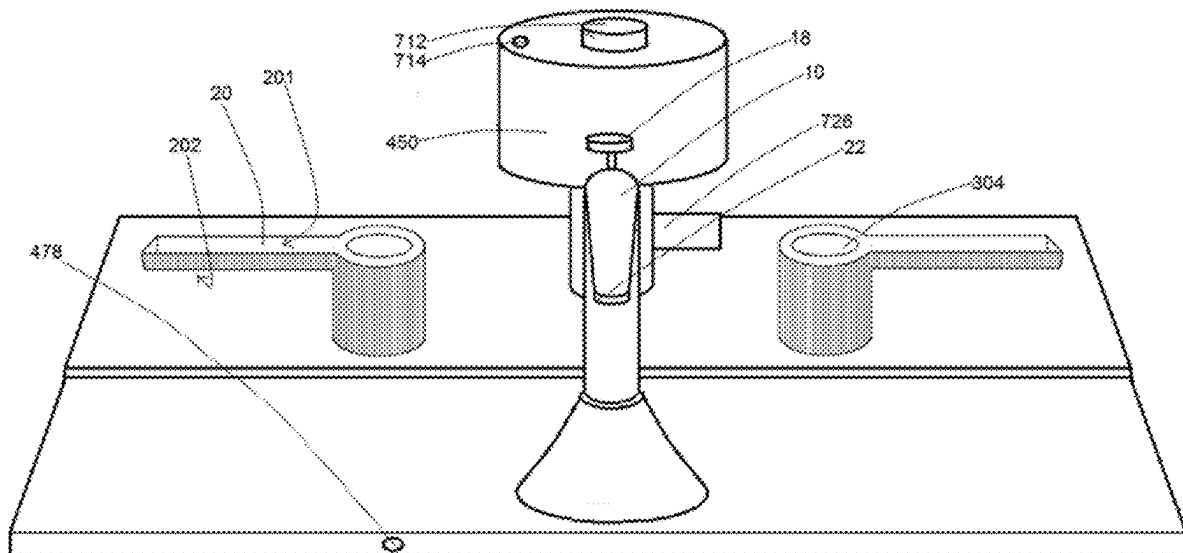
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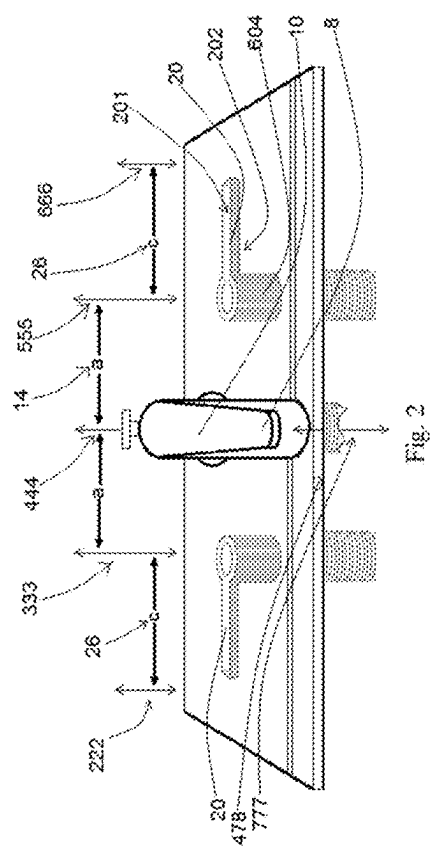
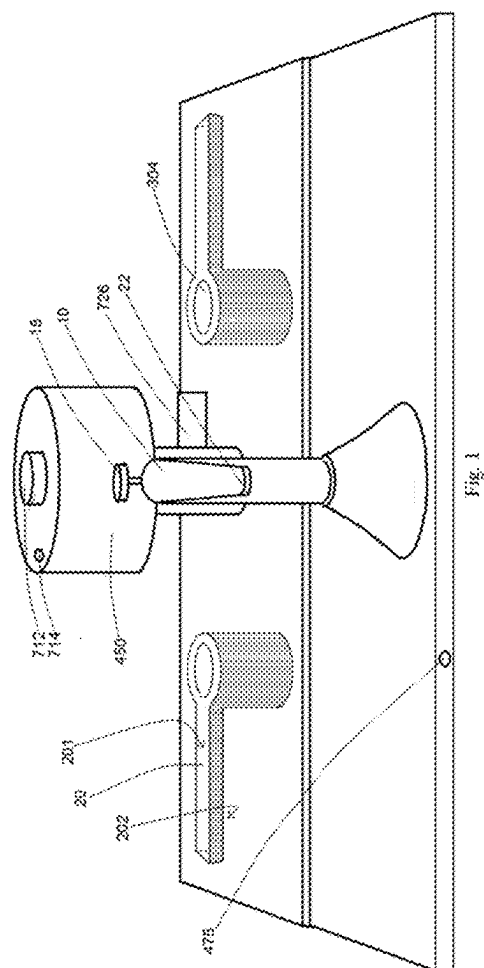
ABSTRACT(21) Appl. No.: **18/704,755**(22) PCT Filed: **Dec. 4, 2023**(86) PCT No.: **PCT/US2023/082393**

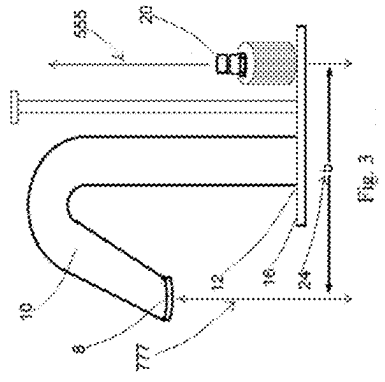
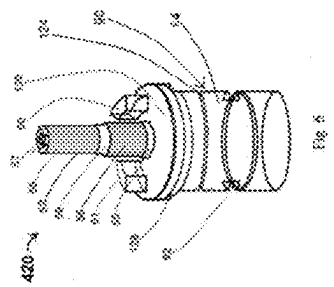
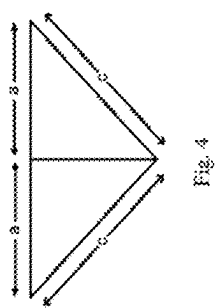
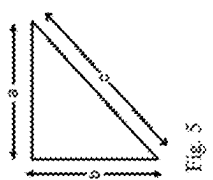
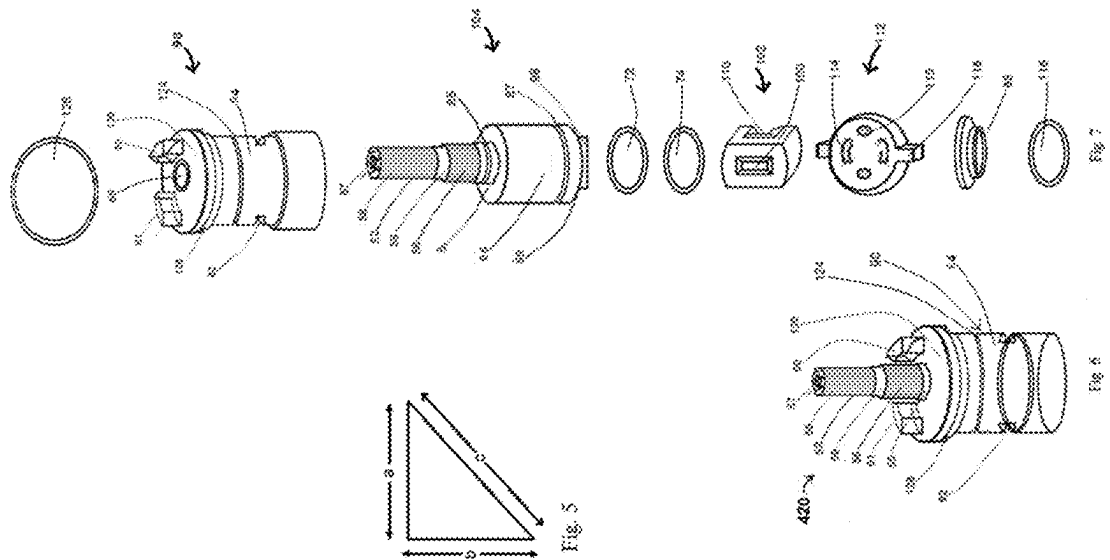
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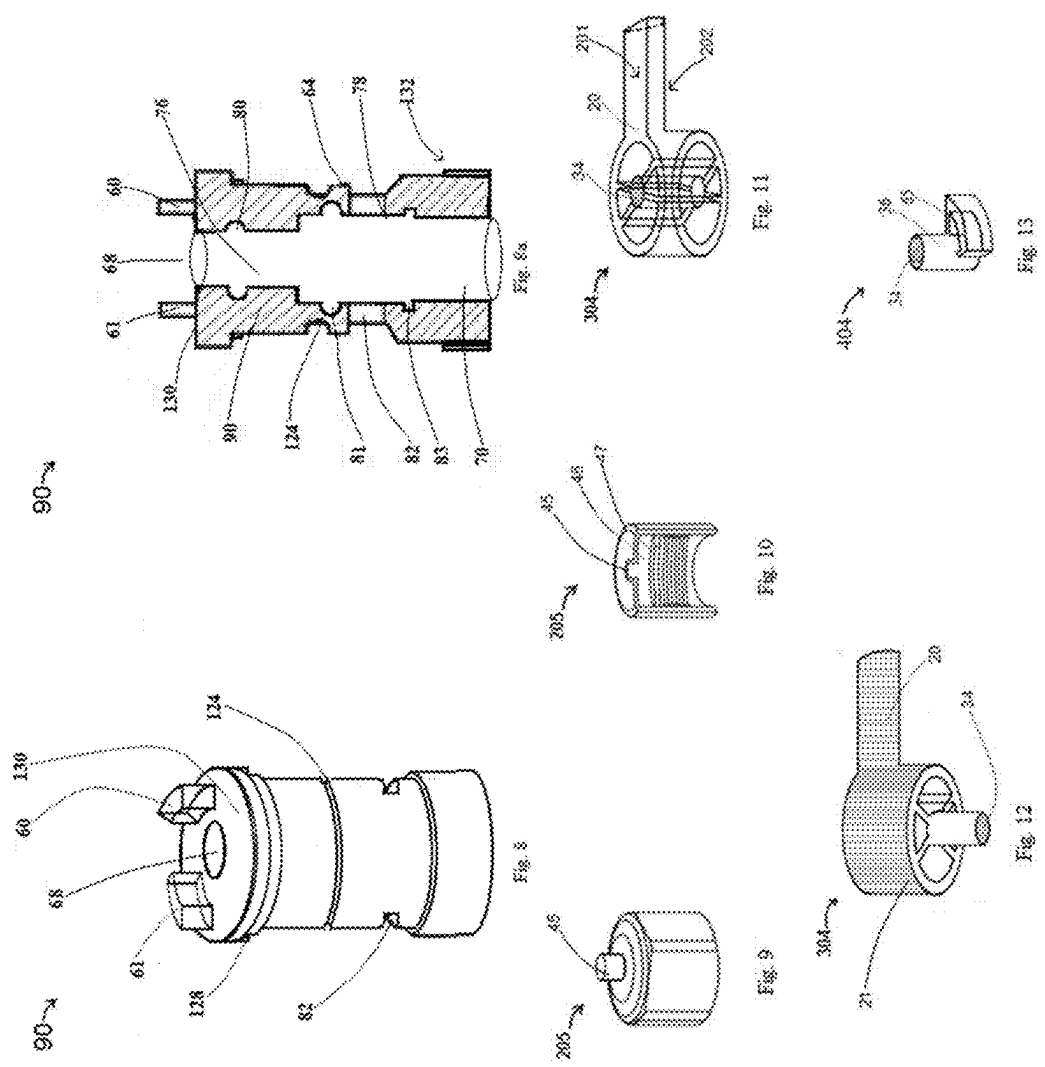
(2) Date: **Apr. 25, 2024****Related U.S. Application Data**(60) Provisional application No. 63/475,718, filed on Dec.
3, 2022.**Publication Classification**(51) **Int. Cl.***E03C 1/046* (2006.01)*A47K 5/12* (2006.01)

A washable faucet lever with crank-activated detergent dispenser having a lever cartridge having a stem operably coupled to a hand lever and a two-bar linkage configured to rotate in a linkage axis of rotation parallel to and simultaneous with a lever axis of rotation, a detergent reservoir member configured to house a liquid detergent therein and that includes a reservoir valve, an encasement member defining an enclosed region therein and an outlet, and having an outlet valve, wherein the dispenser includes a piston disposed within the enclosed region, coupled to the two-bar linkage, and operably configured to translate within the enclosed region upon rotating the hand lever in the lever axis of rotation to selectively and simultaneously displace the reservoir valve and outlet valve to permit flow of the liquid detergent through an internal channel of the reservoir member, into the enclosed region, and out through the outlet.









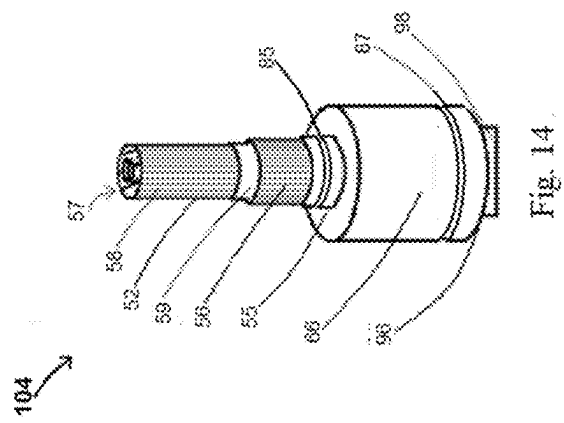
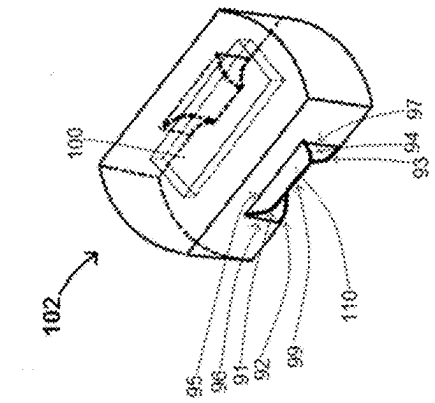
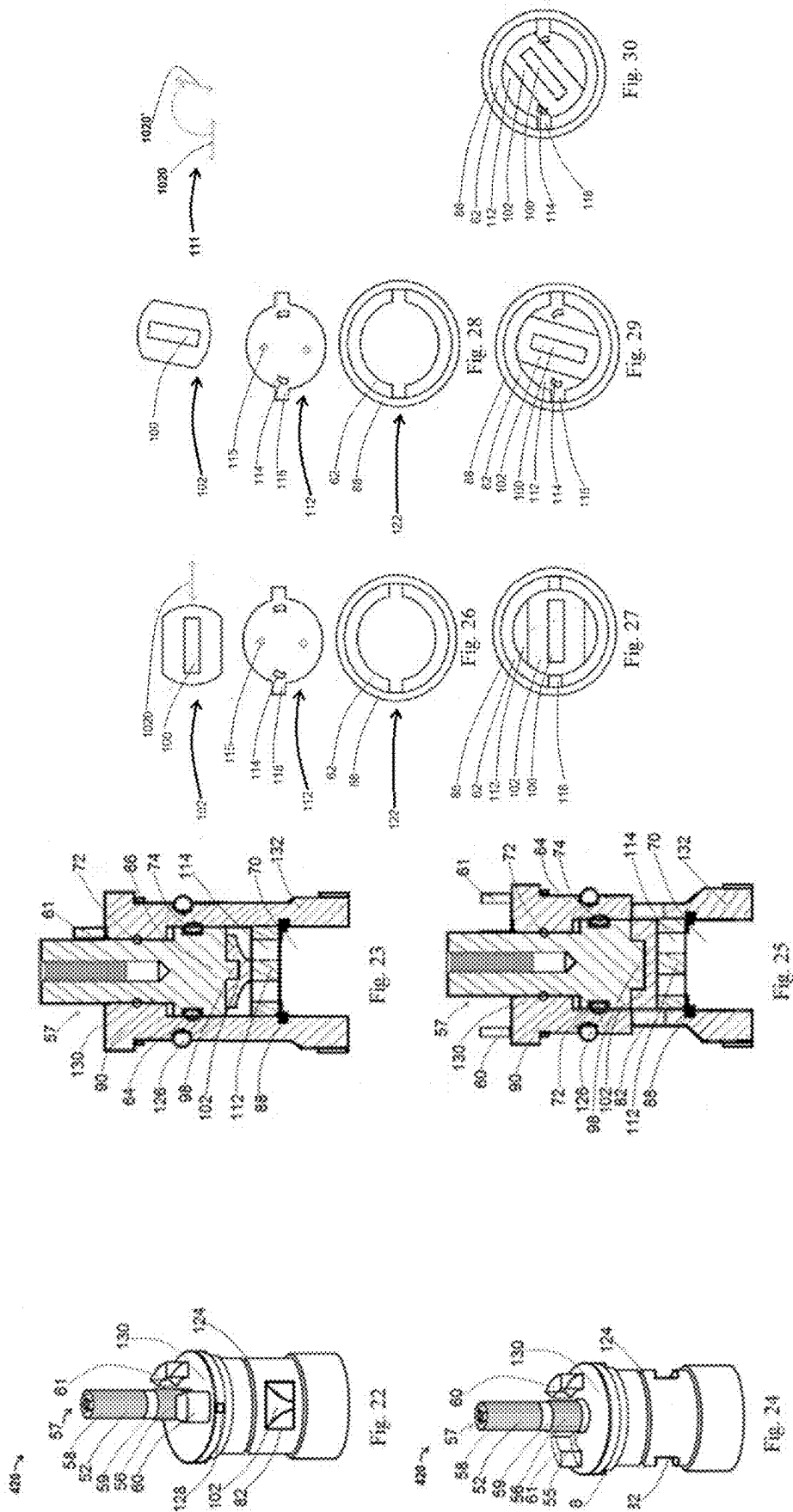
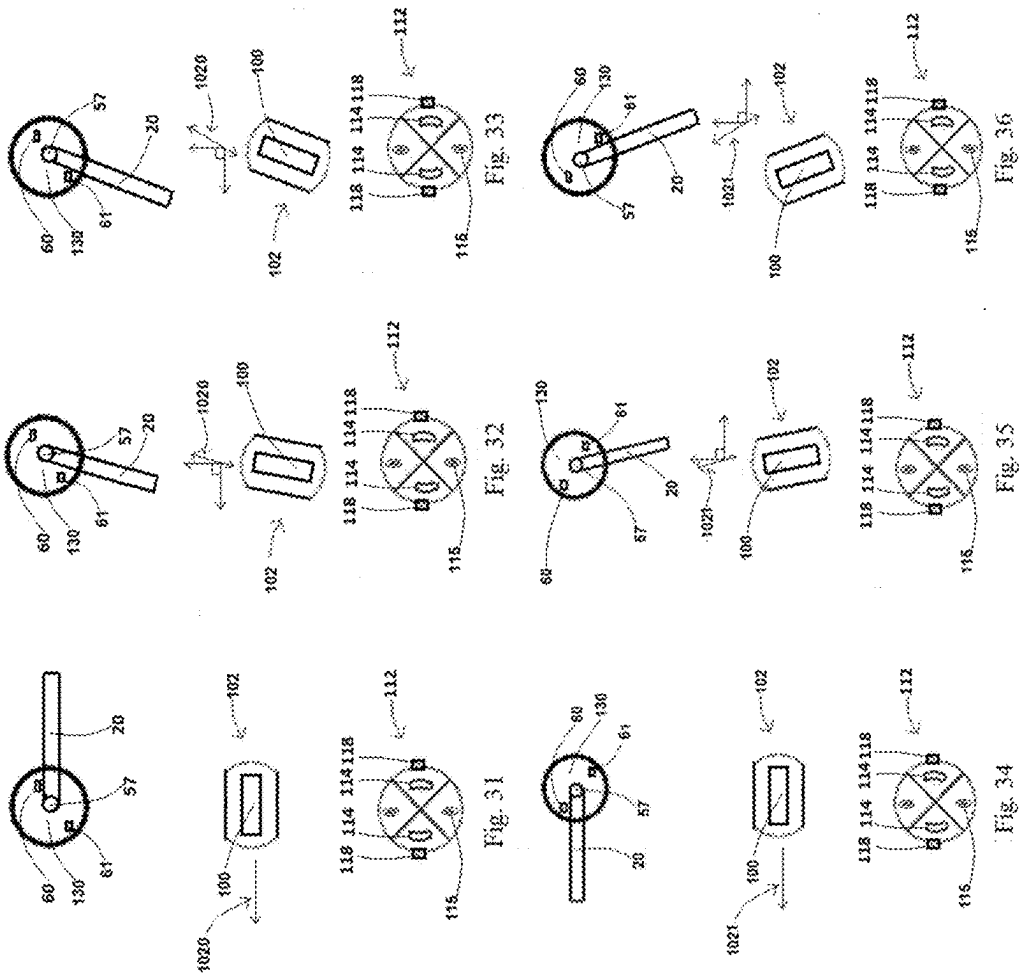


Fig. 14







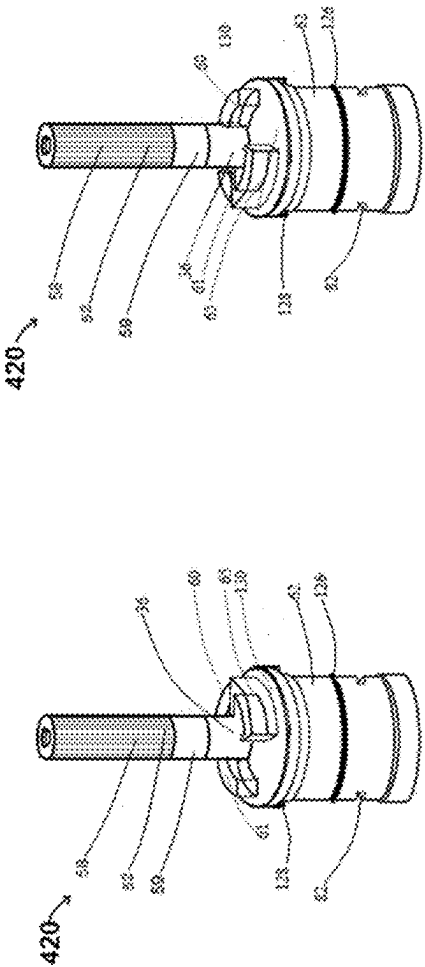


Fig. 38

Fig. 37

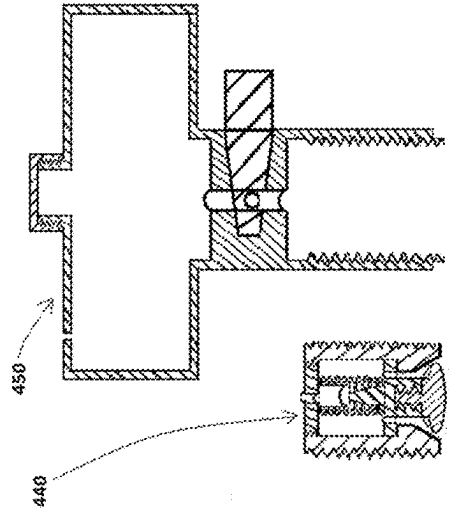


Fig. 40

Fig. 43

Fig. 42

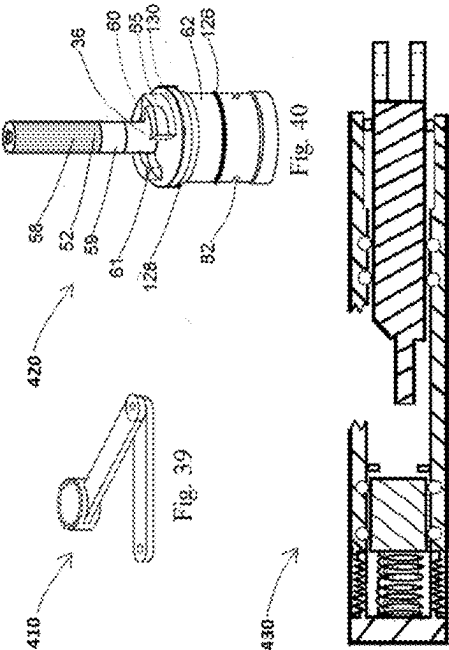
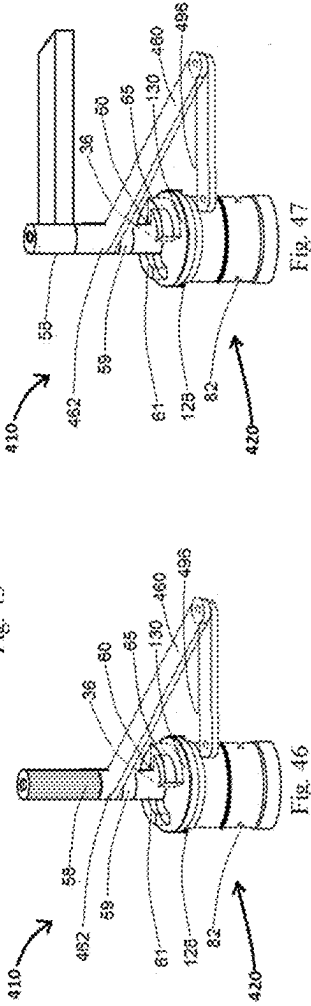
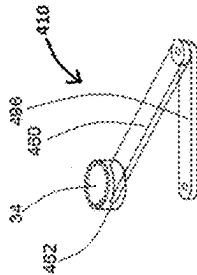
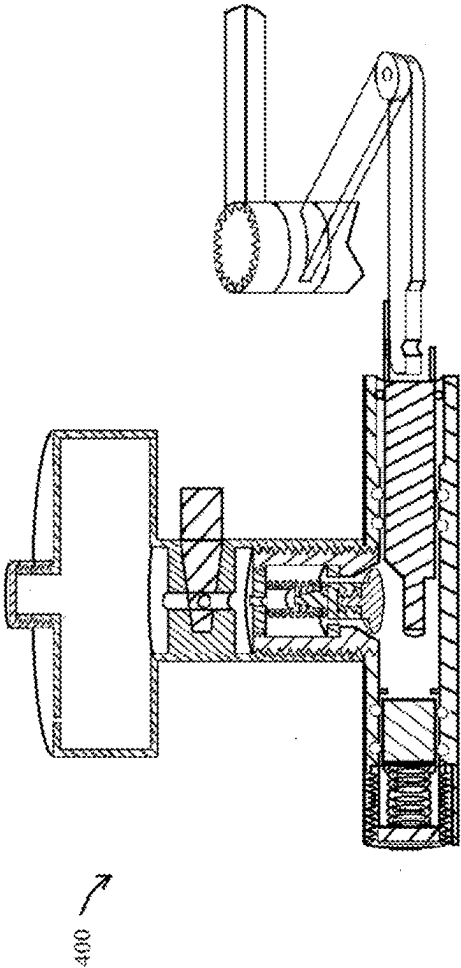
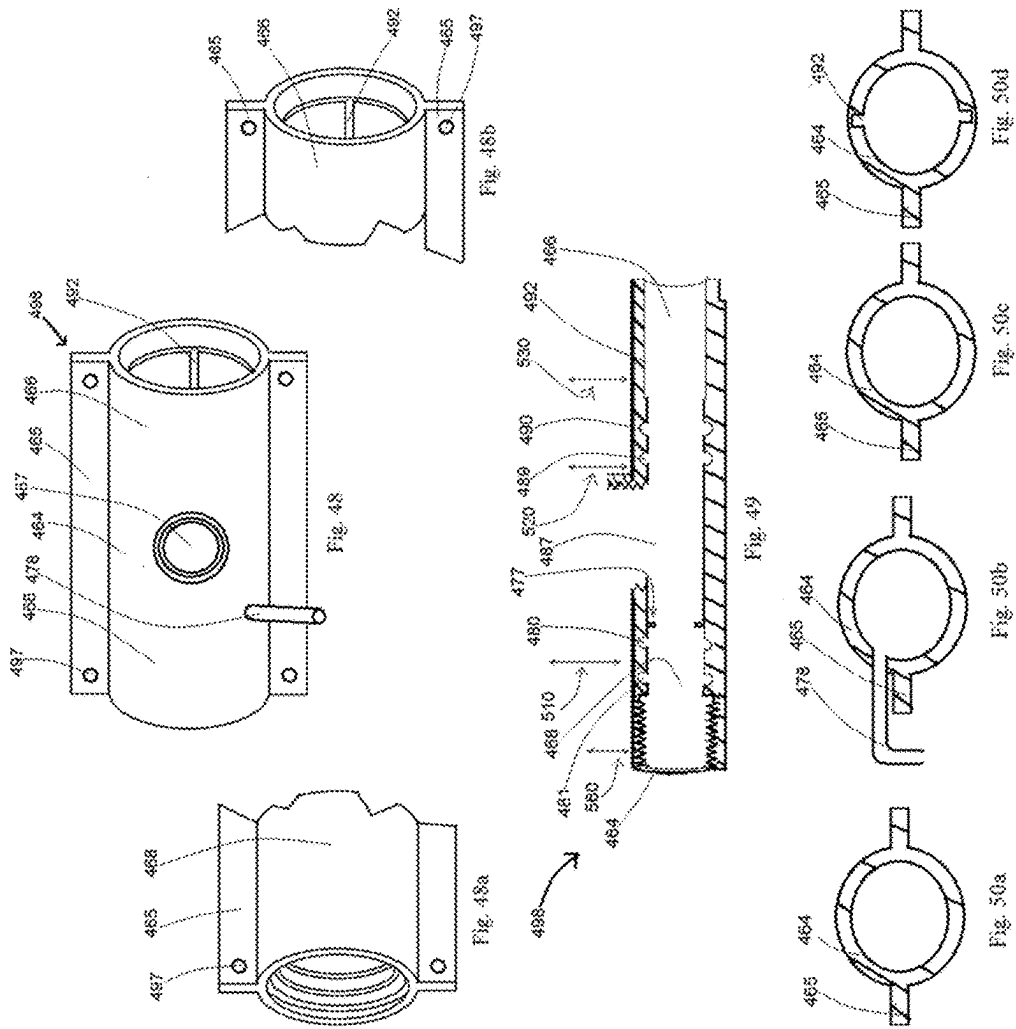


Fig. 39

Fig. 41





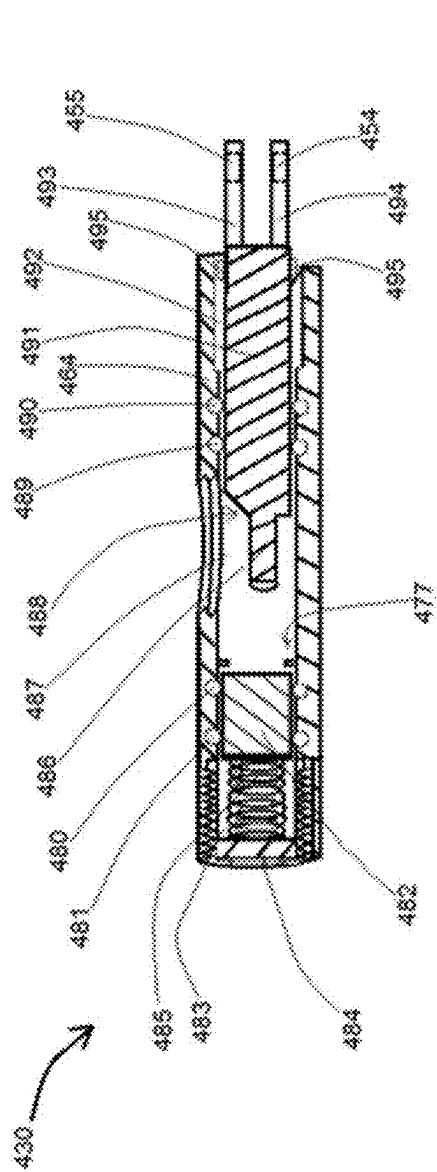


Fig. 51

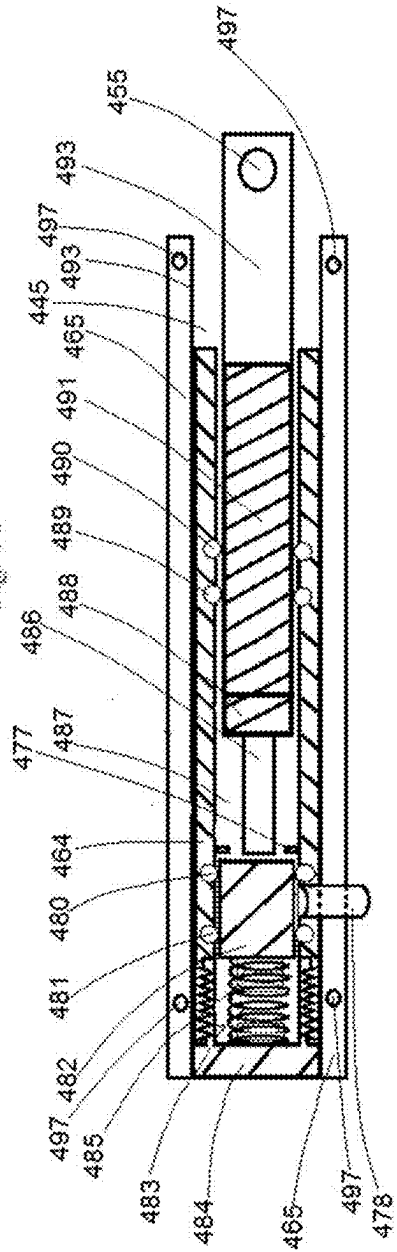
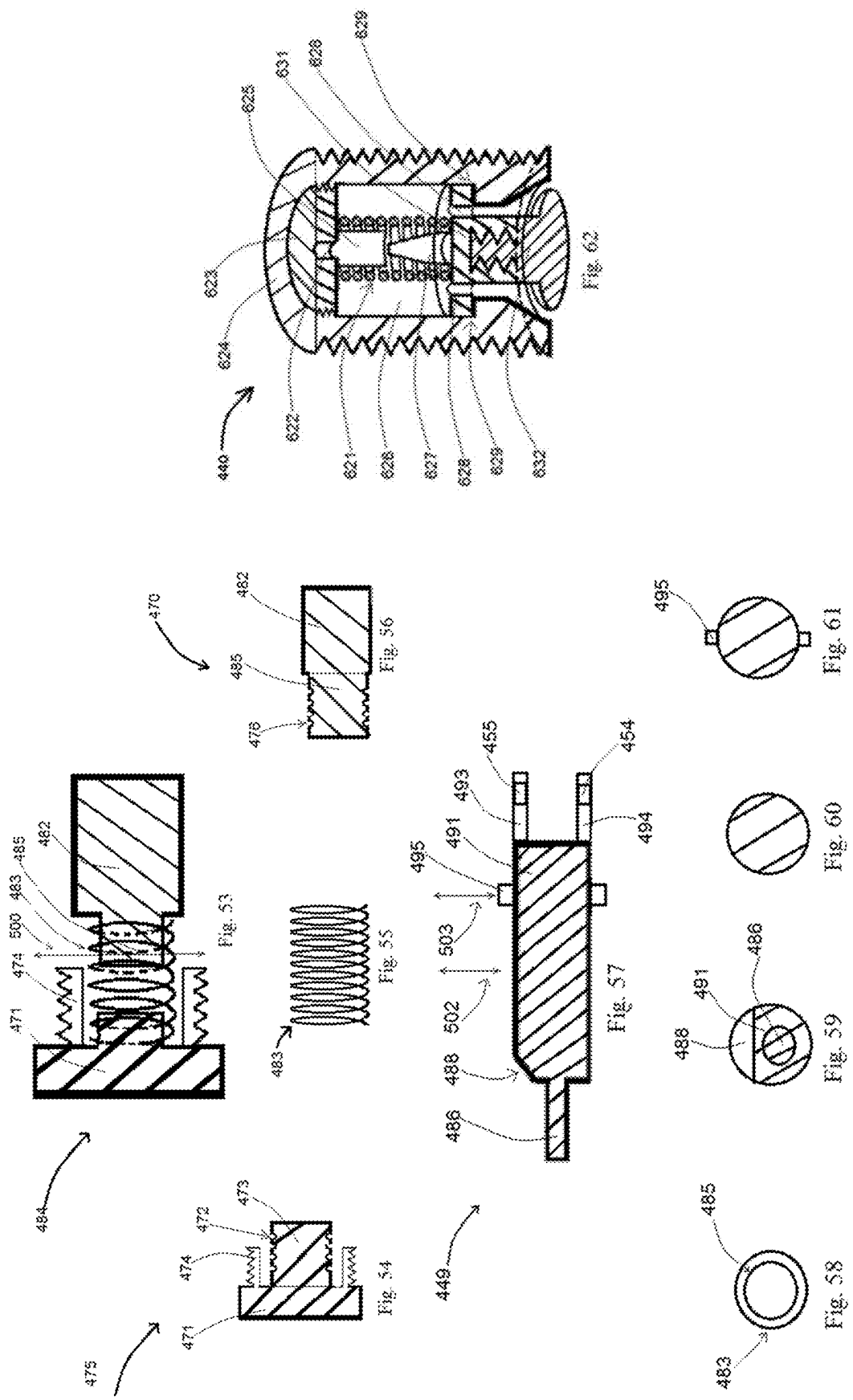
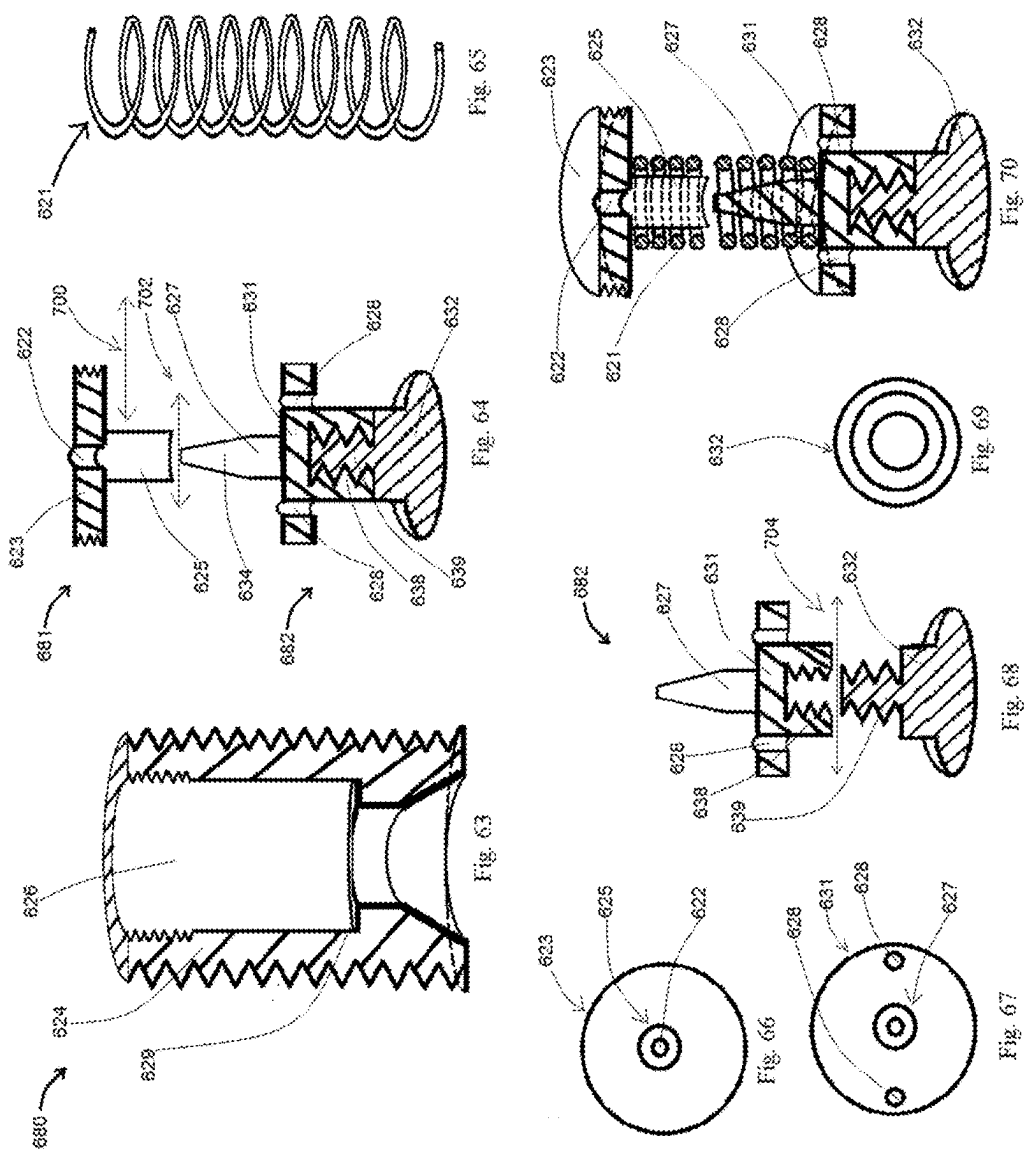


Fig. 52





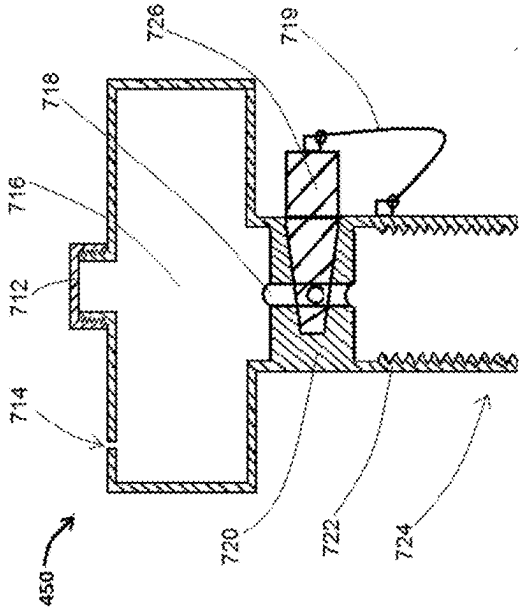


Fig. 71

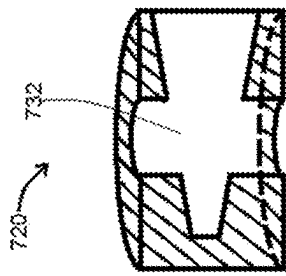


Fig. 72

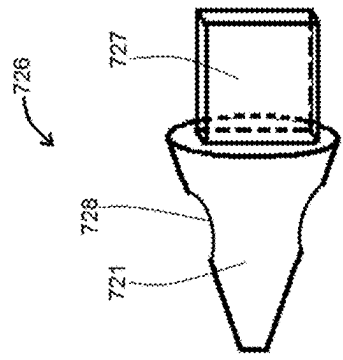


Fig. 73

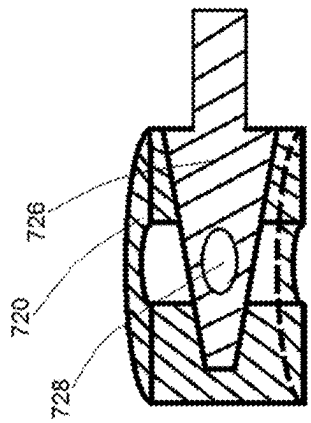


Fig. 75

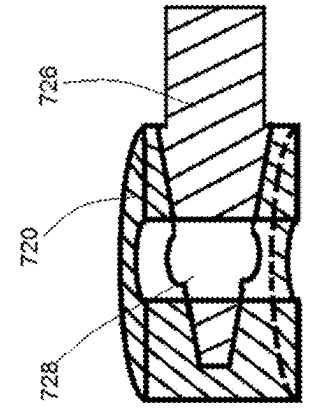


Fig. 76

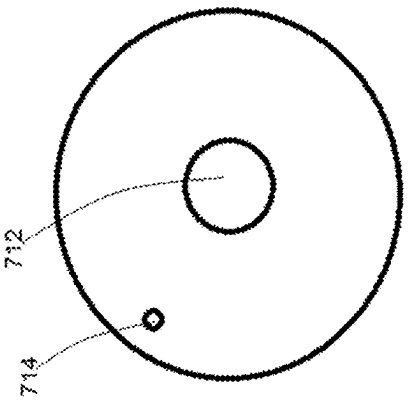


Fig. 74

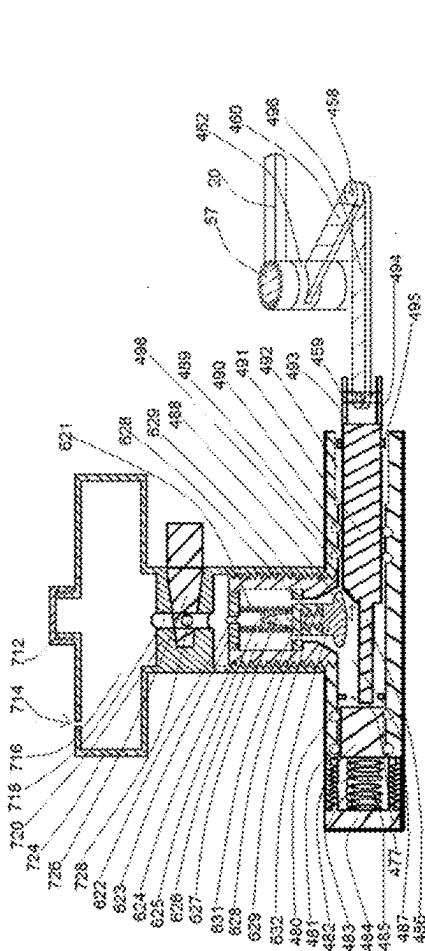


Fig. 77

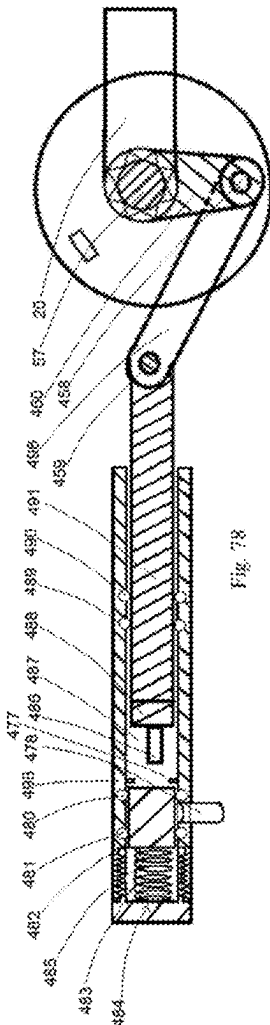


Fig. 78

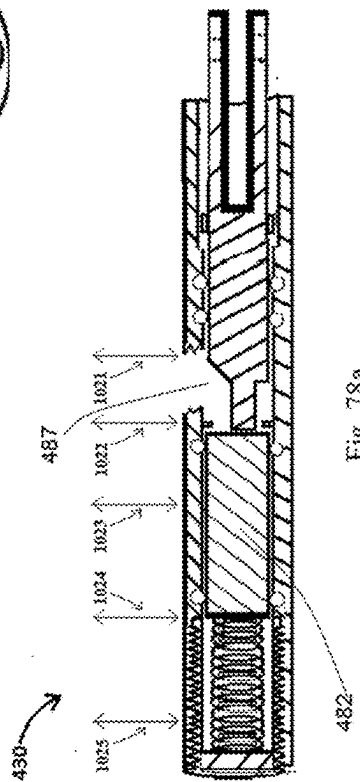
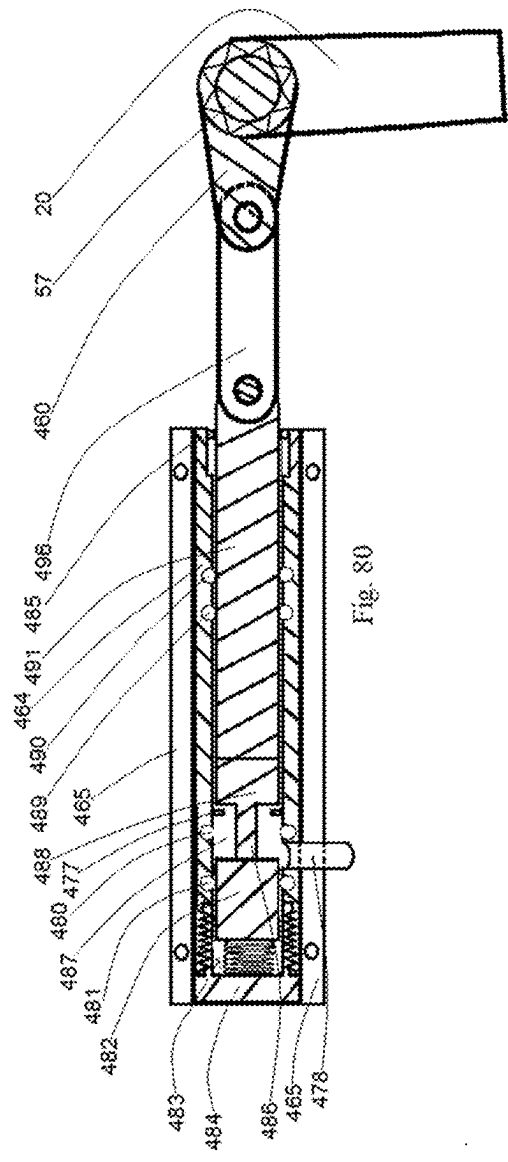
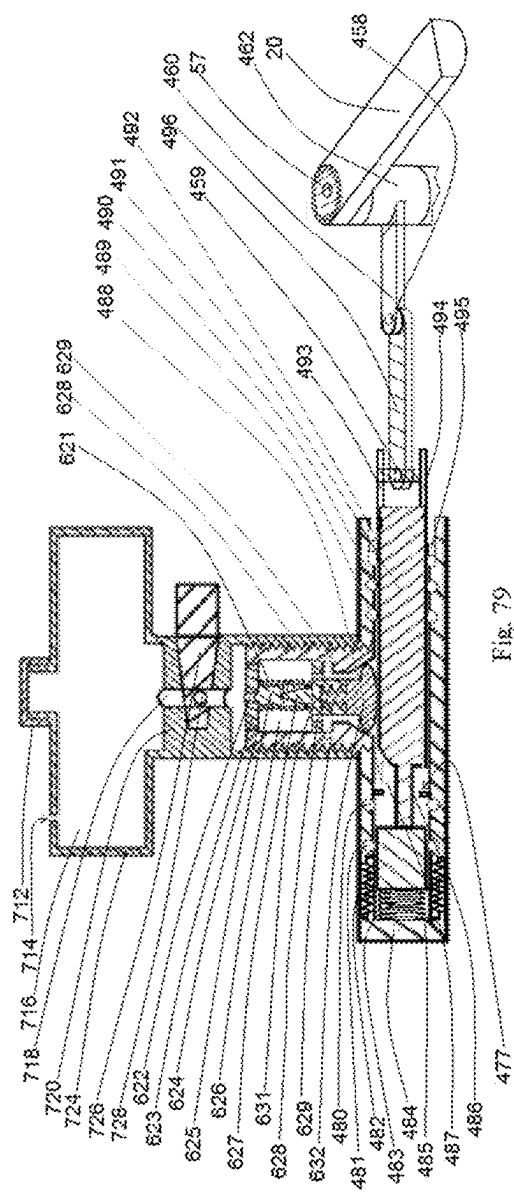
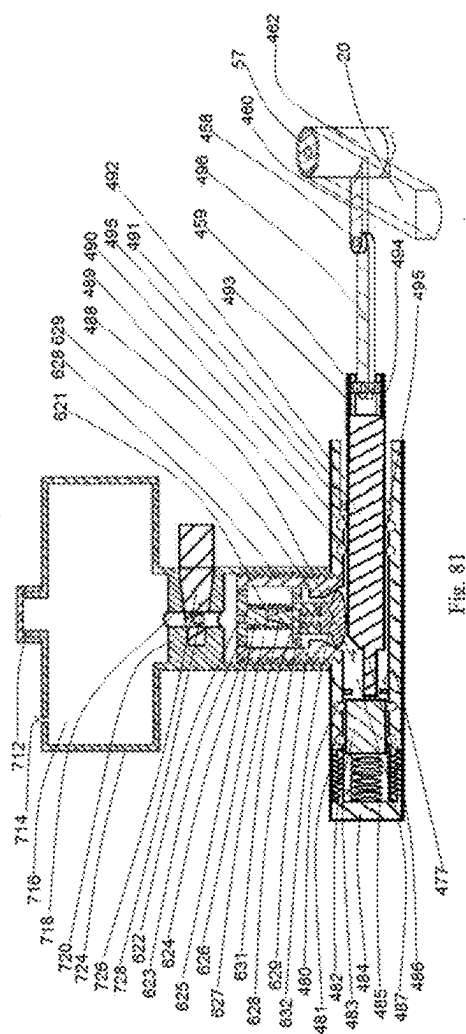
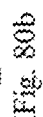
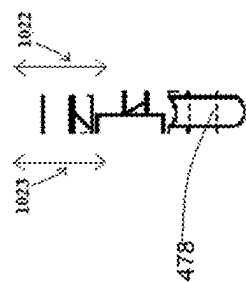
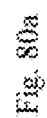
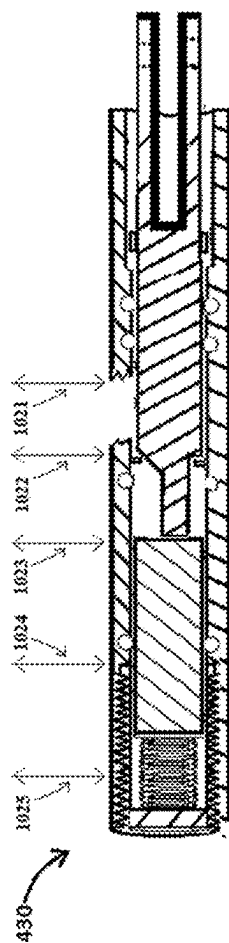


Fig. 78a





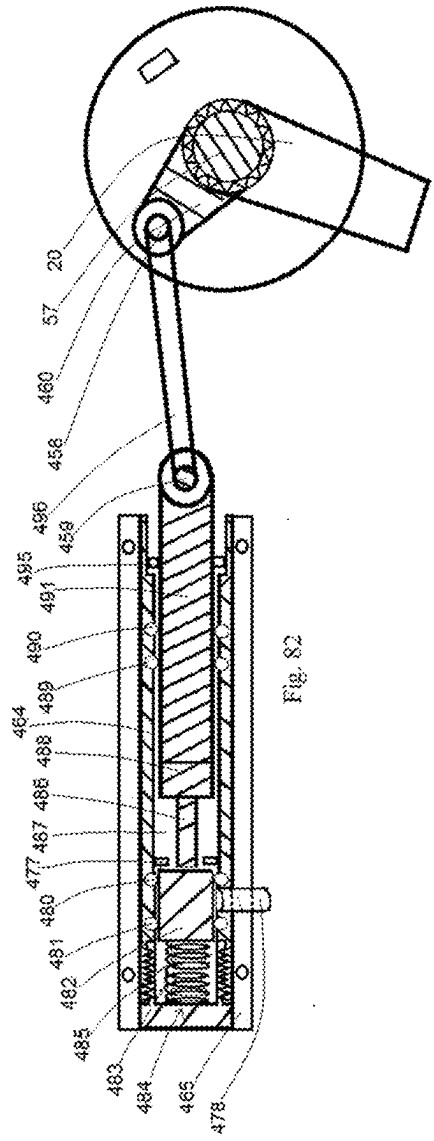


Fig. 82

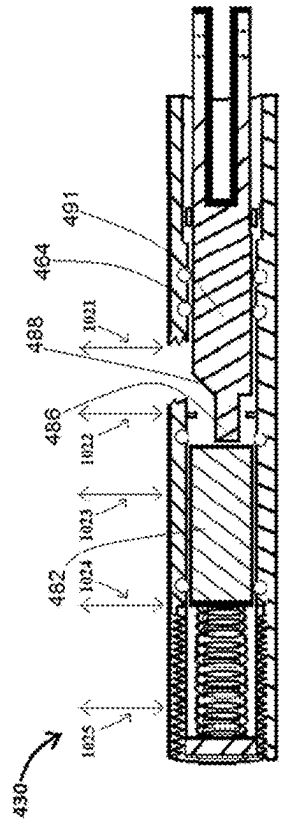


Fig. 82a

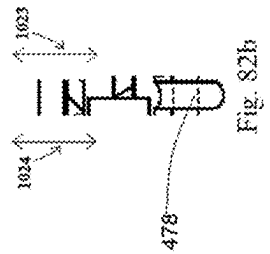


Fig. 82b

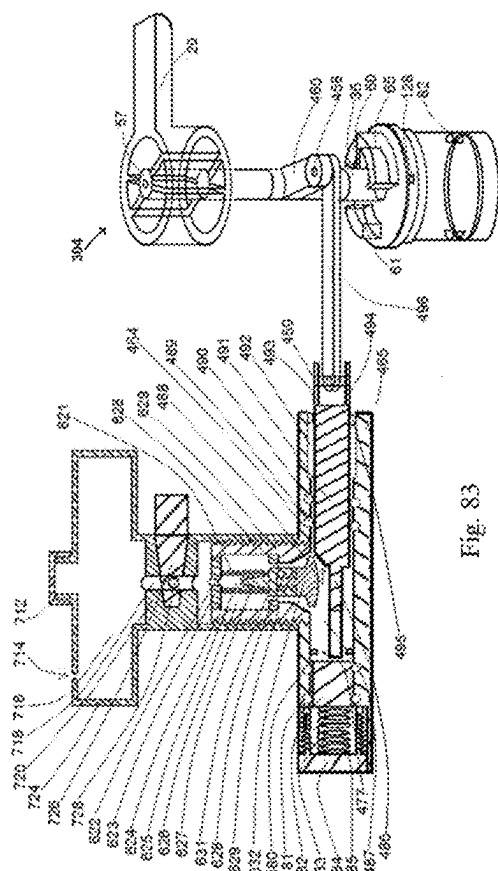


Fig. 83

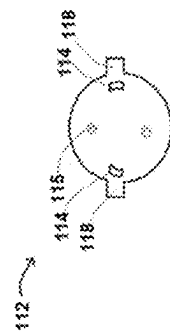
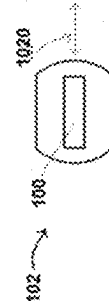
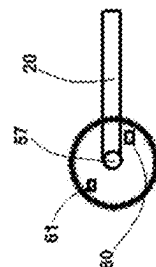
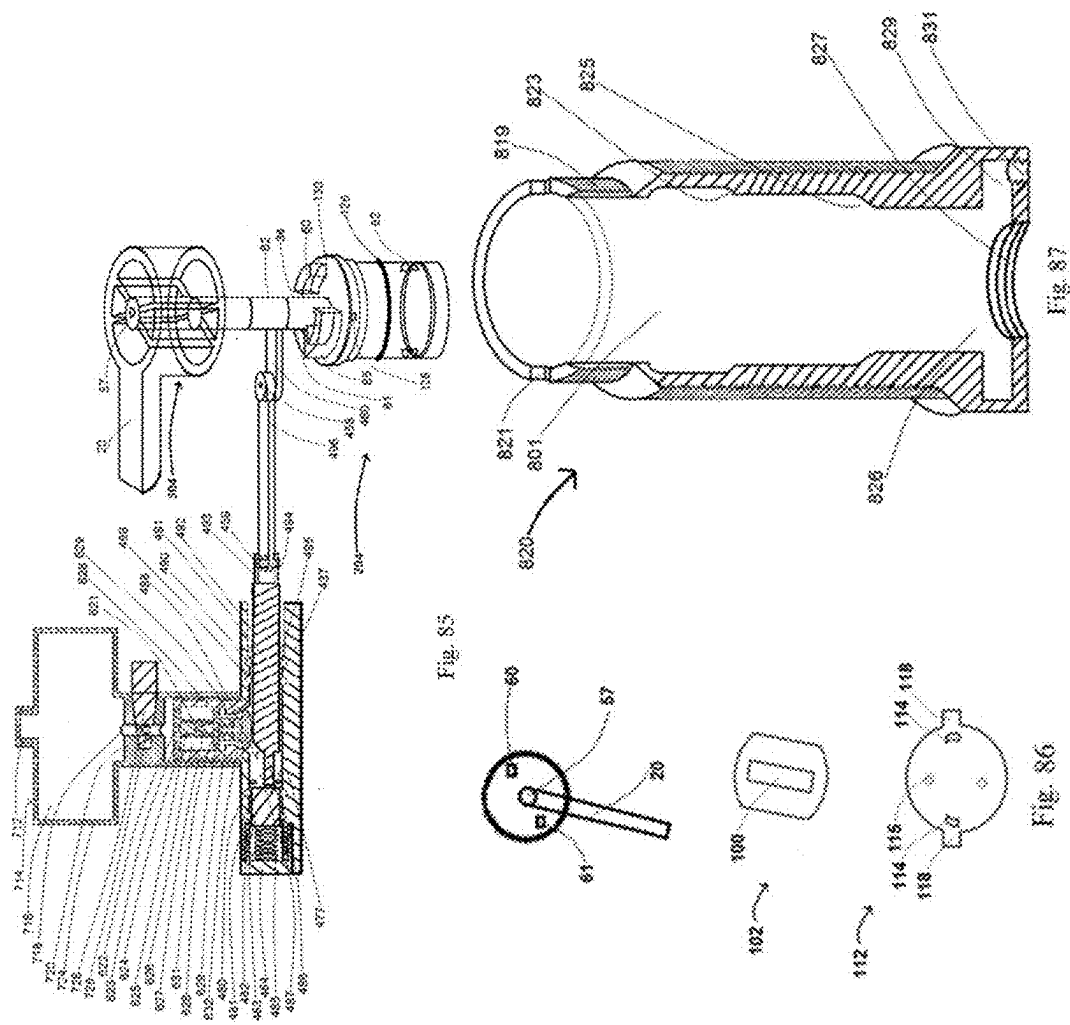
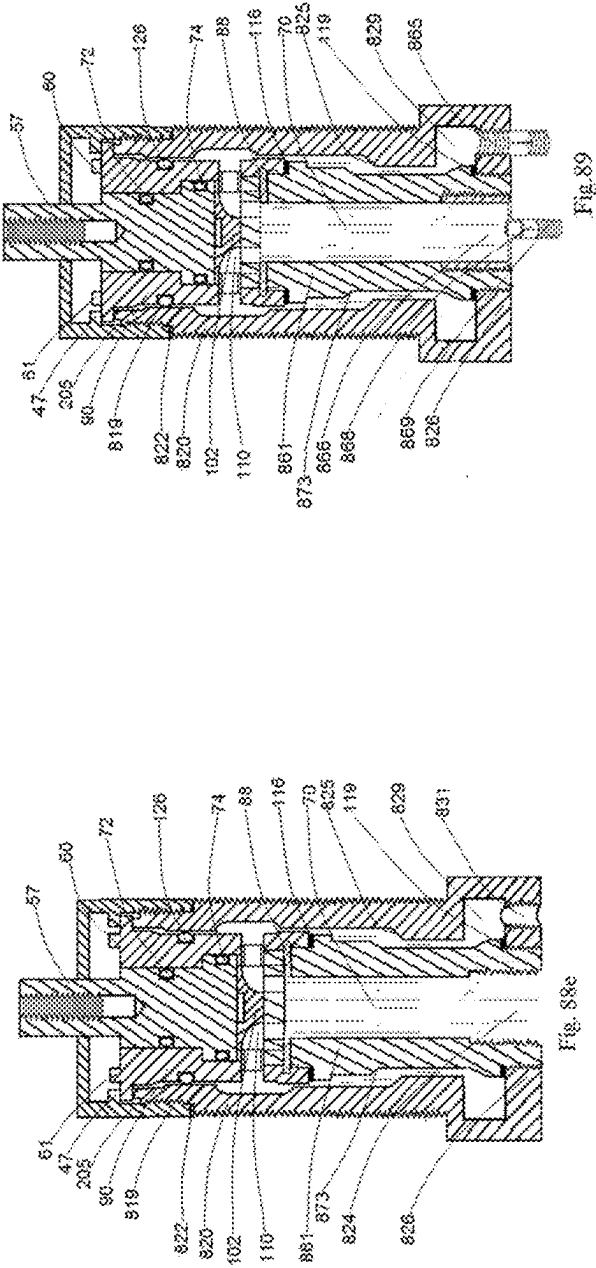
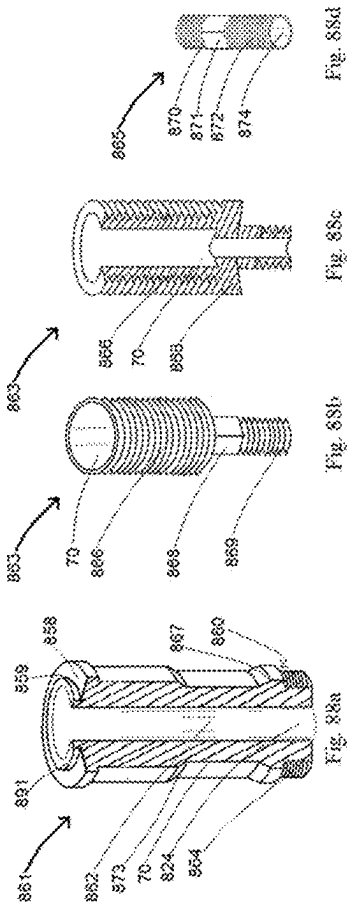


Fig. 84





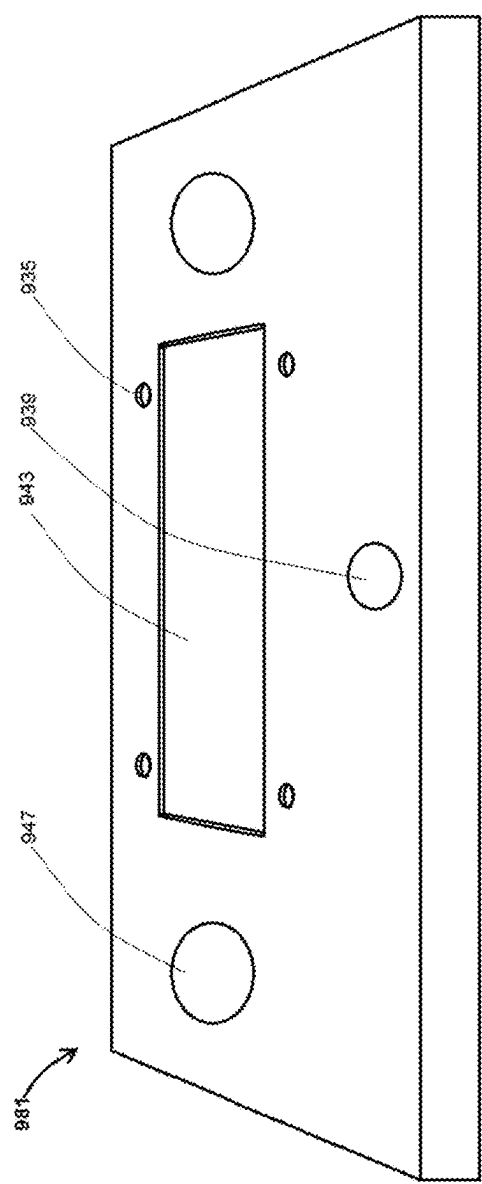


Fig. 90

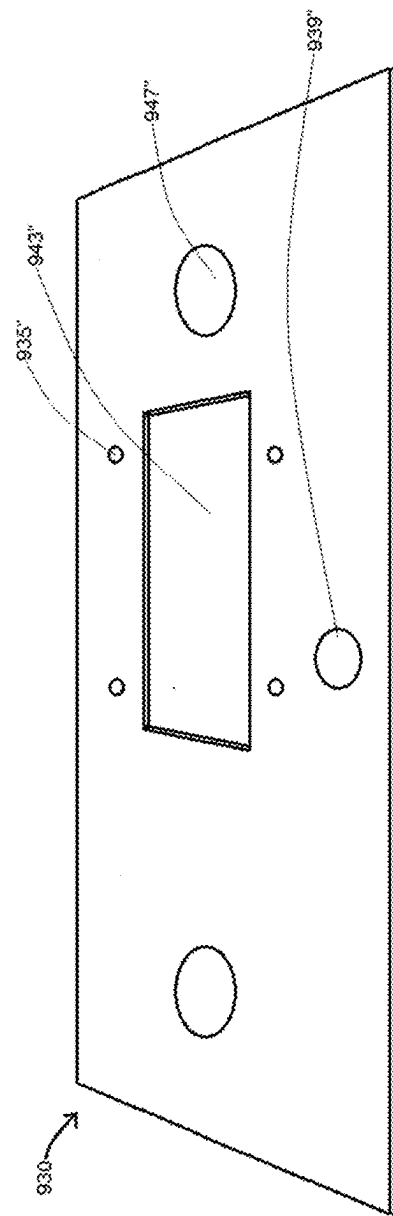
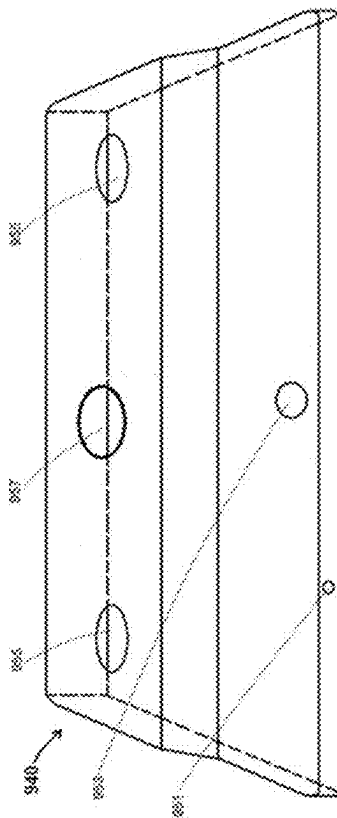
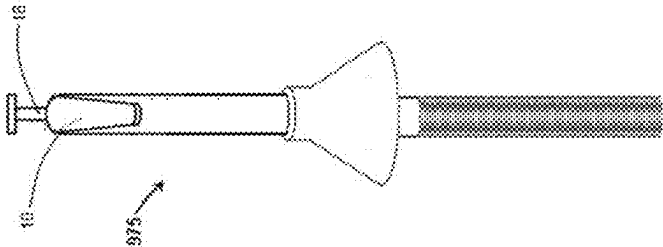


Fig. 91



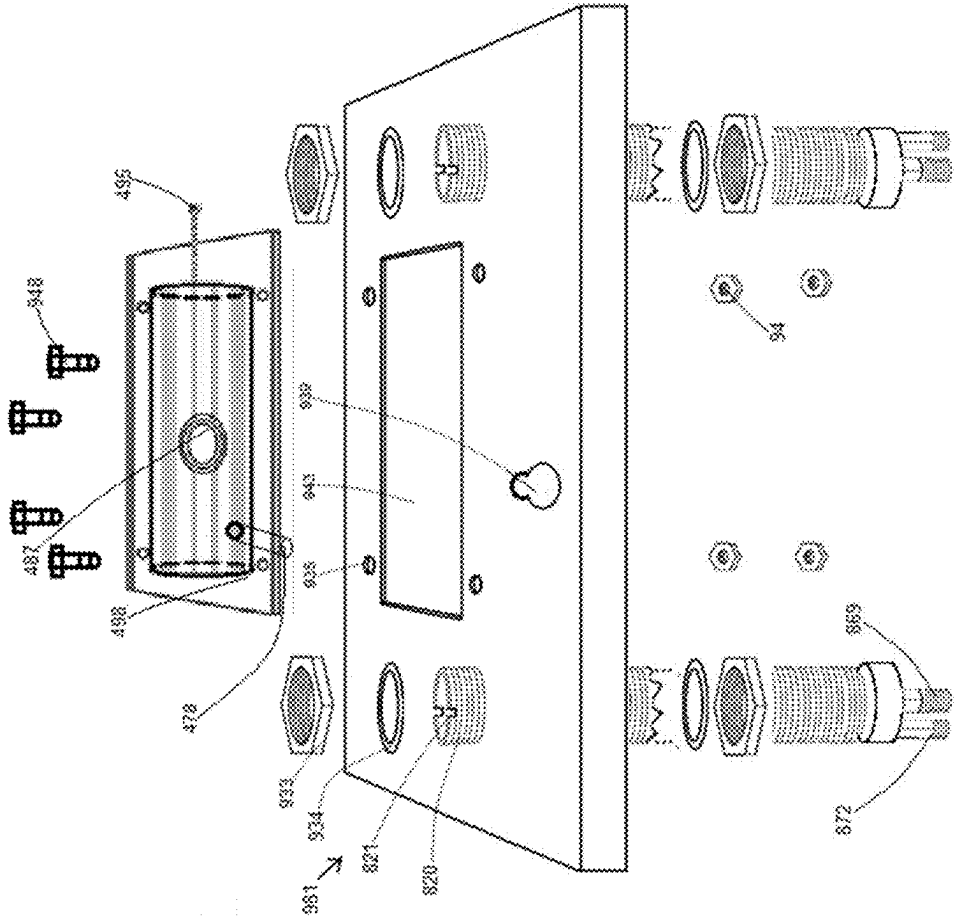


Fig. 93

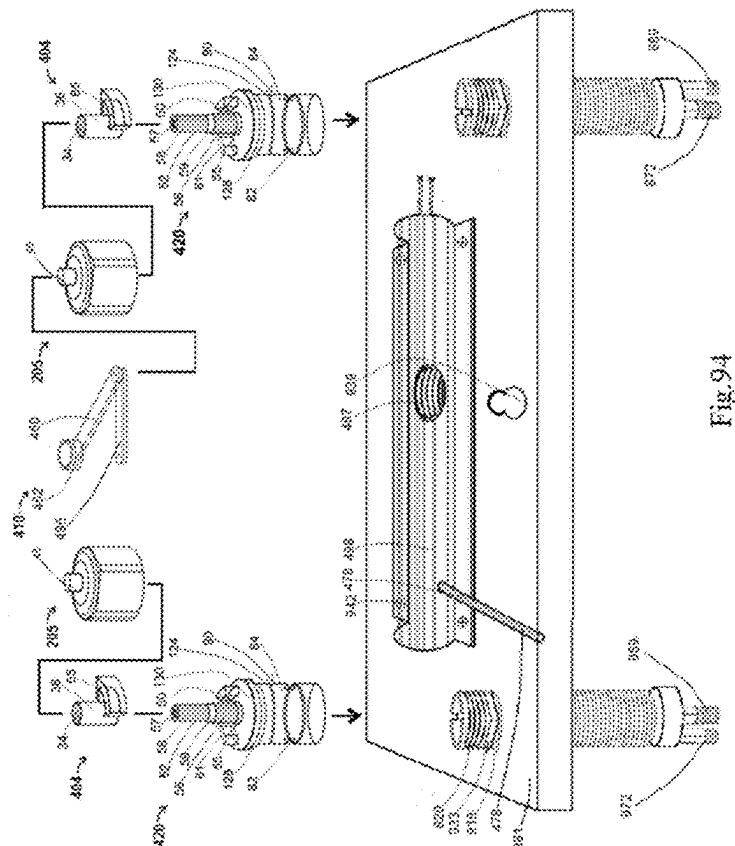


Fig. 94

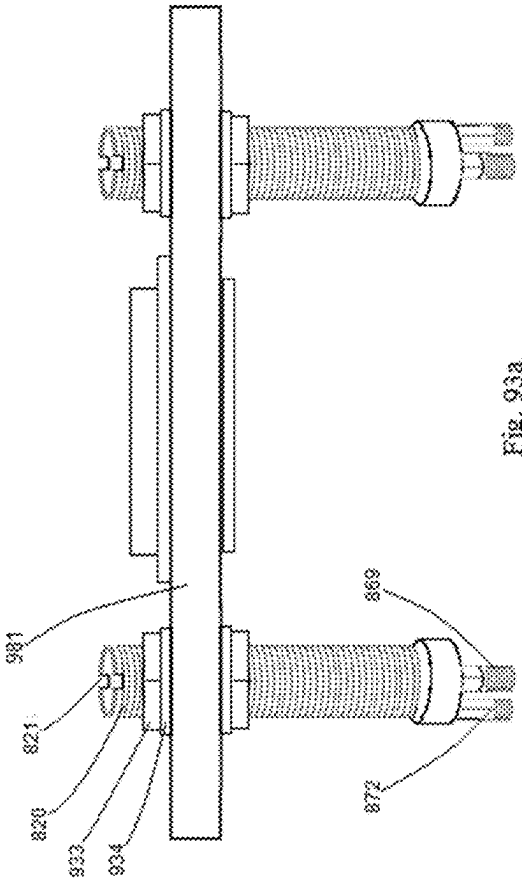
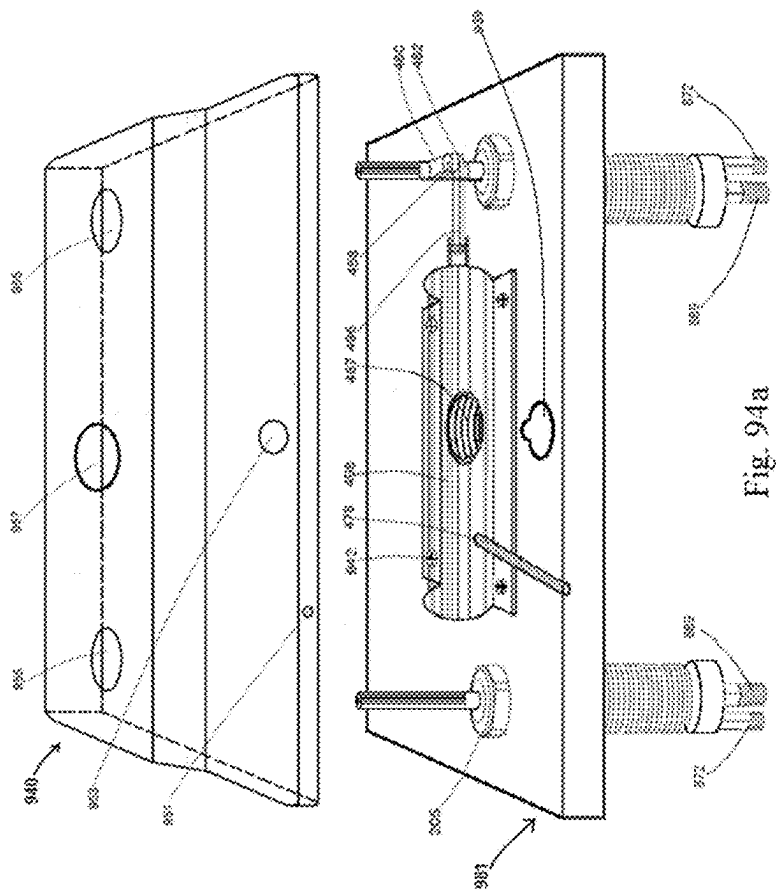
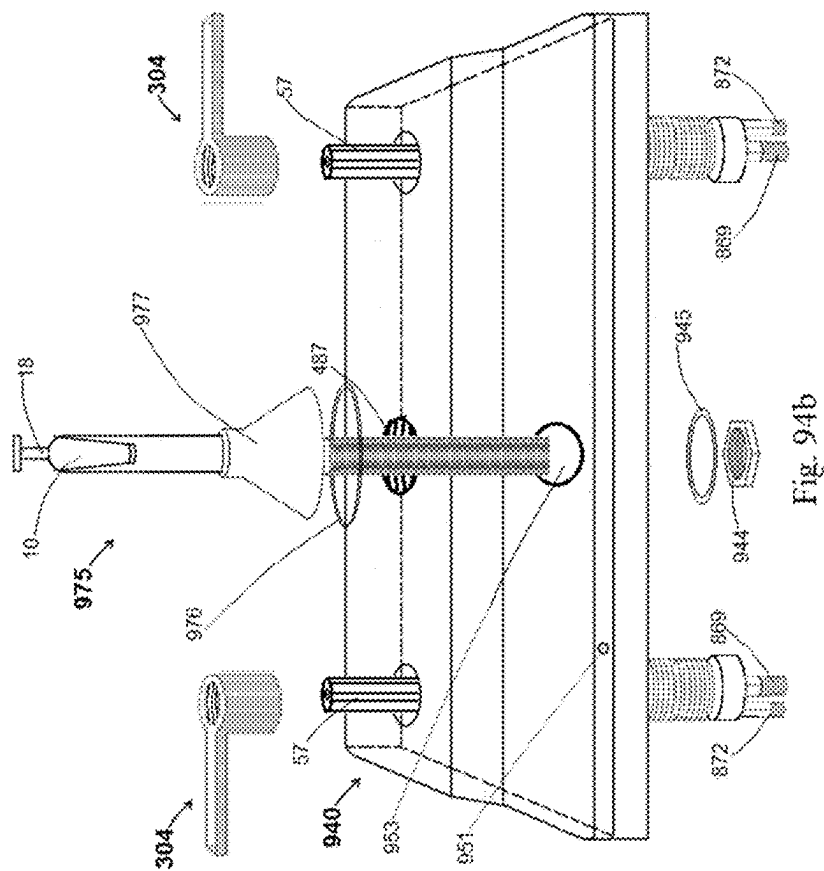
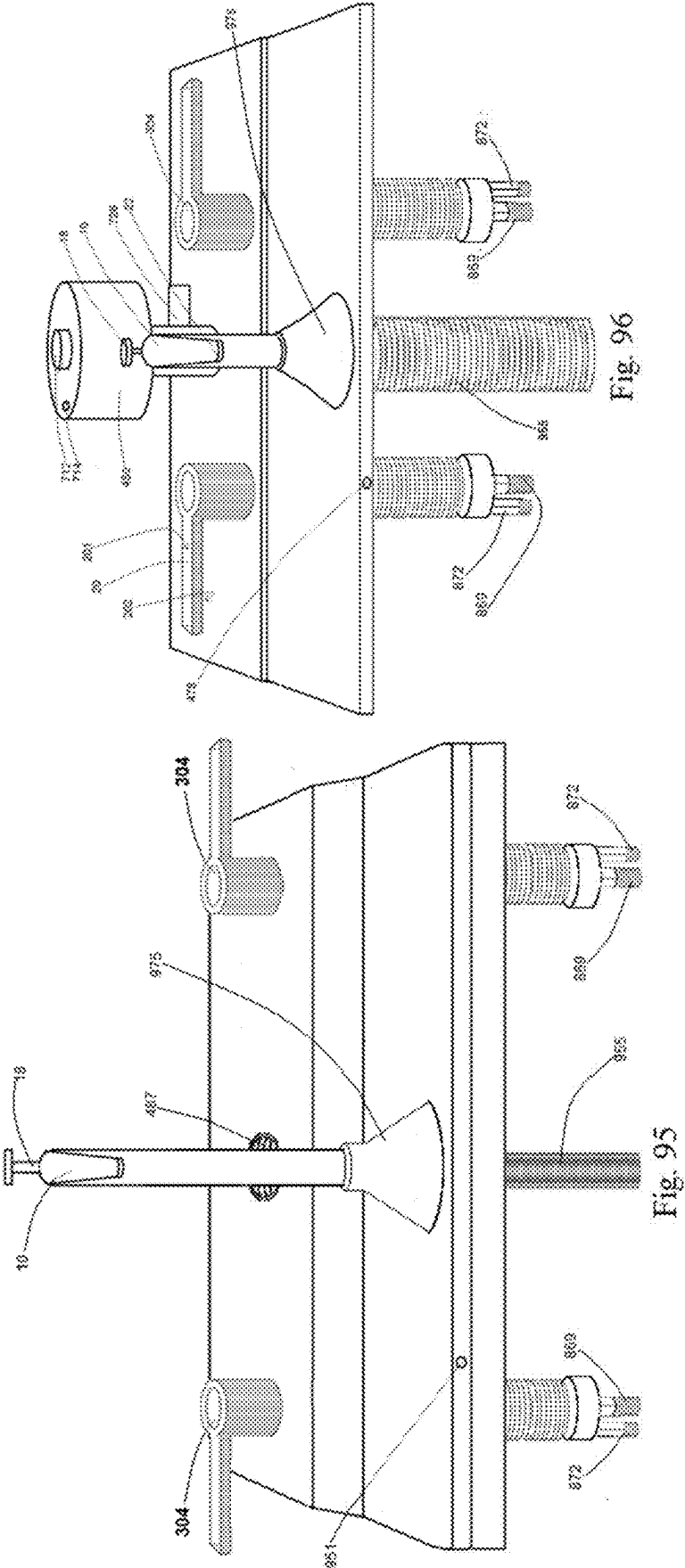
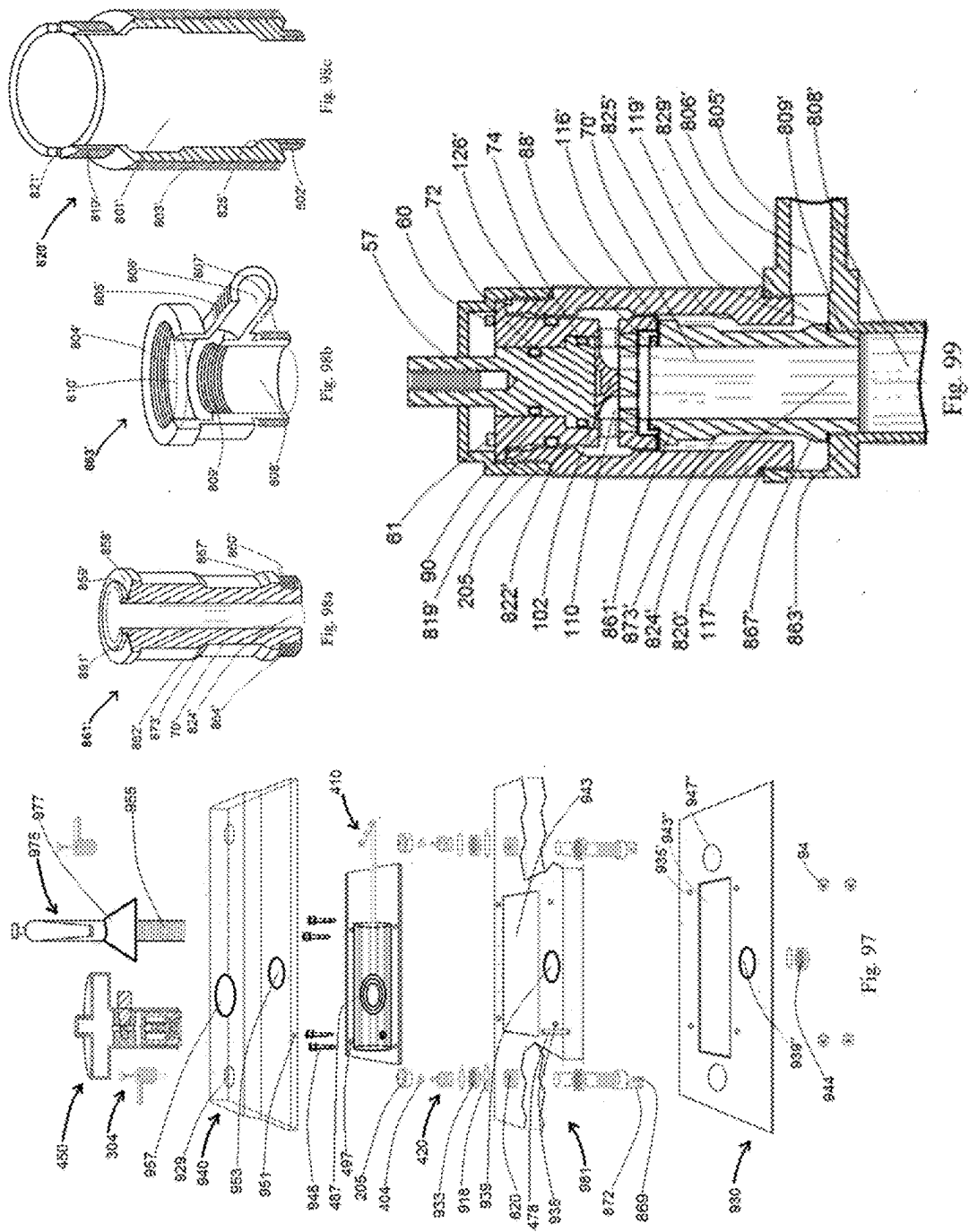
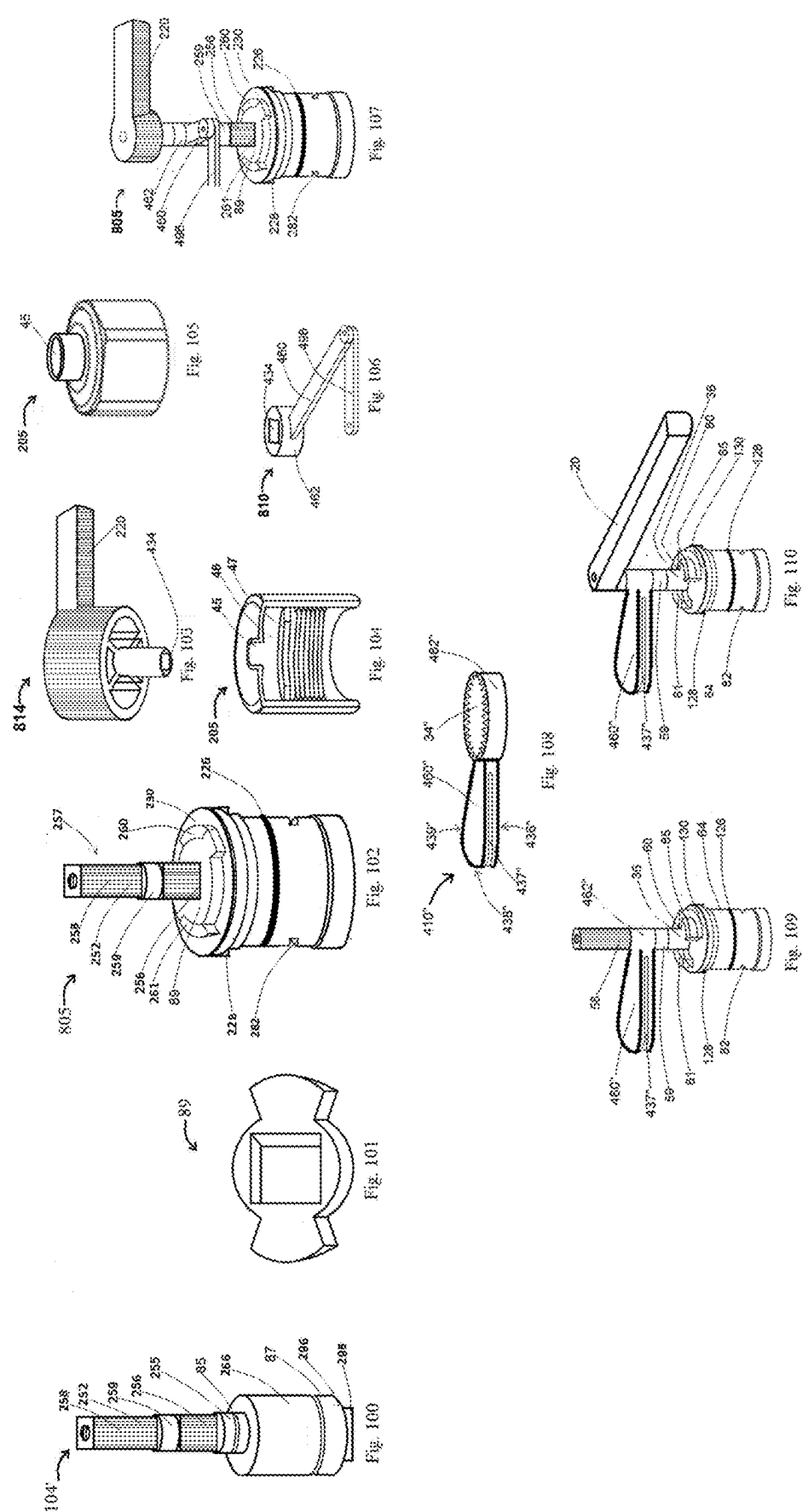


Fig. 93a









**WASHABLE FAUCET LEVER WITH
CRANK-ACTIVATED DETERGENT
DISPENSER**

CROSS-REFERENCE

[0001] This application claims the benefit of provisional application Ser. No. 63/475,716 filed on Dec. 3, 2022 by the present inventor.

BACKGROUND TO INVENTION

[0002] The taps of current bathroom faucets are usually activated by handles. Currently, handle-less faucets taps are predominantly located in restrooms of airports among other places. These handle-less faucets taps serve to significantly reduce the spread of contagious diseases that may have a devastating effect on the well-being of human beings amongst other possibilities. These faucets however are not economical, additionally such handle-less faucets have a limited lifespan when compared to lever enabled faucets.

[0003] Traditional manually operated faucets in homes are usually fitted with cartridge valves requiring a handle for their activation; consequently, everyone in a home or school or locations where faucets with handles are used is exposed to being infected by handle enabled faucets. Currently, handle enabled faucets require that a lever be turned for activation, however the spout is set apart from the handle; this is so because one is able to get water running by turning a lever handle through at most a right angle for existing handle enabled faucets.

[0004] Activating the lever can contaminate the lever handle with pathogens. After hands are washed for example, these same hands again are used to activate the lever in order to turn off the flow of water, consequently hands or other parts of the body may again be contaminated by the residual pathogens that initially were left on the lever before hand washing. Current faucets are not designed to wash the lever or dispense detergent or antibacterial solution when a faucet lever is activated.

[0005] Many designs of manually operated faucet valves exist; however, there is a need for a manually operated faucet enabled with lever washable detergent dispensing capabilities. Furthermore for reasons of economy and human wellbeing such a lever washable detergent dispensing faucet should be affordable. Additionally, such a design of a lever washer detergent dispensing faucet valve should be in keeping with creative modern faucet design features, be long lasting while being disposed to manipulation by a consumer.

[0006] Additionally, the advent of globalization facilitates the rapid spread of diseases that could significantly impact world health and economies on large scale; such diseases previously were localized and thus did not pose a threat to the global economic or well being of humanity. It is conceivable that there are unknown pathogens and that there will always be the threat of life-threatening pathogens at this point of global human development and beyond.

[0007] Electronic eye faucets or handle-less faucets are available, for example, as seen in Stauffer, U.S. Pat. No. 4,767,922, entitled a Hand Presence Activated Water Faucet Controller, or in Chen, U.S. Pat. No. 9,347,207, entitled Faucet Assembly. However, these faucets are not generally affordable by the global population in general. Consequently, a faucet system that inherently enables a showering

of the lever and dispensing hand washing detergent while being affordable thus mitigates the spread of pathogens in schools, homes and other institutions and facilities where people share the use of faucets.

[0008] Furthermore, by integrating a lever activated soap dispenser with a washable lever significant reduction in the transmission of pathogens can be achieved.

BRIEF SUMMARY

[0009] According to an embodiment of the disclosed subject matter, the invention provides greater protection against the transmission of pathogens such as the Covid-19 virus, H1N1 virus, flu virus and other transmittable pathogens when hands are being washed. The invention includes a lever handle enabled valve cartridge that efficiently operates while dispensing detergent and enabling its' lever handle to be washed when hands are being washed.

[0010] The invention includes a stem assembly that functions to enables the lever handle to be washed by water flowing from the spout of a bathroom faucet tap utilizing preferably a valve cartridge; this is enabled by appropriately dimensioning the length of the lever in relation to the positioning of the spout and faucet cartridge of the invention thus enabling lever to intercept the downward direction of flow of water from the spout of a faucet using preferably a cartridge valve.

[0011] The stem also initiates a detergent dispensing unit to dispense detergent from a detergent dispensing unit while being enabled to reduce the flow of solvent through the spout of the faucet unit when the lever is turned to enable flow of water through the spout of a faucet. The stability of the stem is also enabled by the application of a bonnet nut that stabilizes the stem to a cartridge mount. Embodiments of the invention enables detergent to be dispensed preferably prior to substantial of water from a faucet spout when the lever is initially turned or during timely rotation of the lever as water flows from the spout of the faucet as the lever is turned.

[0012] The faucet assembly is disposed to have at least one valve cartridge, with the assembly comprising a faucet having at least one faucet handle; a fluid conduit influenced by the faucet handle for activating the cartridge valve, a faucet body having an outlet or spout for releasing fluid at an outlet end and an inlet end in fluid communication with the fluid conduit; a base for mounting the faucet handle and faucet body; the base having an upper surface and at least one cartridge comprising a cartridge body having an exterior surface, an interior surface defining a longitudinally extending passage extending from an inlet opening to an outlet opening, and at least one side opening extending transversely through the body from the interior surface to the exterior surface.

[0013] The stem of the cartridge unit is enabled with an upper end portion and a lower end portion. The upper end portion of the stem is enabled with means to rotate in reaction to the turning of a lever. The upper end portion is further enabled with means for initiating activation of a detergent dispensing unit and also with attachments enabling restriction the rotation of the lever to its predetermined maximum angle of rotation. The upper end portion extends outwardly beyond the outlet opening of the body when the stem member is situated in the passage of the body. The lower end portion is cylindrically configured to fit in rotatable relationship with the interior surface of the body when situated within the passage of the body. Additionally, the

lower portion has a bottom surface configured to contact and interact with the upper surface of a moveable disc. The geometric side wall of the lower portion configured to fit within the passage of the body and in rotatable mirrored relationship with the interior surface of the body and with an inwardly extending sidewall portion defining at least one fluid receiving area wherein the receiving area is configured so that it can be rotated to align with the side opening in the body for allowing fluid flow out of the side opening of the body.

[0014] The fluid receiving area is somewhat parabolic in features and bordered by the surface of a fixed disc, the lower surface of the stem and one cartridge body sidewall opening.

[0015] The receiving area is configured so that it can be rotated to align with the side opening in the body for allowing fluid flow out of the side opening of the body; a stationary disk having at least one opening, but preferably two symmetrically arranged there through for allowing passage of fluid from an inlet opening of the body through the side opening of the body when the opening in the stationary disk is aligned with the fluid receiving area in the moveable disk, the stationary disk having a side wall capable of interacting with the interior surface of the body; and a sealing ring for stabilizing and retaining the stationary disk and moveable disk within the passage of the body, the sealing ring designed simultaneously capable of mating with the body and interfacing with a conduit to ensure stability and reliability for the interacting stress of its function.

[0016] The exterior surface of the body preferably comprises at least one sealing groove for receiving a sealing component such as an o-ring. The exterior surface of the body may also be enabled with means for seating the valve cartridge within or on a fluid conduit; for example, the feature may be a locking tab and the body may comprise two diametric locking tabs on. The upper surface of the cartridge body may be enabled with means for limiting rotational movement of the valve cartridge when in use; for example, there may be two raised preferably rectangular prismatic shaped stop features on a circle of the upper body surface. The position of the raised stop features a function to restrict the rotation of the lever to its maximum predetermined angle of rotation. The lower end portion of the stem may have a groove on the exterior for receiving a seal, such as an o-ring, between the interior surface of the body and the lower end portion of the stem. The interior surface of the cartridge body may also be enabled with a groove near the inlet end of the body for receiving a sealing ring such as an o-ring. The sealing rings may preferably be configured from suitable flexible and elastomeric material such as silicone or nitrile rubber or appropriately compatible allied sealing intervention such as fluoro-elastomers.

[0017] The stationary disc and moveable disc are preferably configured from ceramic. The stem preferably is made from metal, preferably stainless steel or brass. The upper end cartridge body is preferably made from copper, brass, or stainless steel.

[0018] The cartridge body preferably has two side openings, and the stationary disk preferably has two openings for allowing passage of water from the inlet opening of the body then through the side openings of the body when the openings in the stationary disk are aligned with the fluid receiving area in the moveable disk. The cartridge body is preferably made from copper, brass, stainless steel or poly-

meric or hard rubber composite such as polyether ketone, polyphenylene sulfides, Bakelite or allied compatible hard composite rubber or plastic.

[0019] Preferably, there are two fluid receiving areas in the moveable disk, and the stationary disk is enabled with locking mechanism that is capable of mating with a mating groove on an exterior surface of the body thus ensuring restrictive rotary motion. The preferably two opening access on the stationary disk is further oppositely aligned to accommodate the function of the lever assembly; the function of the lever assembly is to ensure that the lever can be enabled for washing while ensuring convenient location of the lever relative to the function and positioning of faucet parts when not in use.

[0020] The relational distances between the base of the faucet spout and the base of the spindle handle a , the distance between the vertical line of fluid stream from the faucet spout and the base of the spout b and the distance between the vertical line of fluid stream and the base of the faucet handle c satisfy the relationship $\sqrt{a^2+b^2} \leq c$.

[0021] The said washable lever handle initiate flow of solvent by activating rotation of the spindle thus a cam attached to the upper end of the stem will rotate when the lever is rotated, hence by attaching a crank and connecting rod to the lever where such a crank influences a follower with connection to a piston where the piston functions as a projection retraction activating unit for dispensing detergent, significant reduction in the transmission of pathogens can be achieved.

DESCRIPTION OF DRAWINGS

[0022] The accompanying drawings, which are included to provide further understanding of the disclosed subject matter, are incorporated in and constitute a part of this specification. The drawings also illustrate embodiments of the disclosed subject matter and together with the detailed description serve to explain the principles of embodiments of the disclosed subject matter. No attempt is made to show structural details in more detail than may be necessary for a fundamental understanding of the disclosed subject matter and various ways in which it may be practiced.

[0023] FIG. 1 shows an embodiment of a two-handle embodiment of the invention as seen when a finished unit of the invention is mounted and available for utilization.

[0024] FIG. 2 shows a positioning between the lever, the spout, the cartridge and the base of the spout of a finished unit of the invention. The detergent reservoir 450 and a spout escutcheon 977 are not shown in FIG. 2.

[0025] FIG. 3 shows a right-side view of that of FIG. 2.

[0026] FIG. 4 shows a geometric relation for the positioning of the lever, the spout, the cartridge and the base of the spout for a finished two handle unit of the invention.

[0027] FIG. 5 shows a geometric relation for the positioning of the lever, the spout, the cartridge and the base of the spout for a single handle unit of the invention.

[0028] FIG. 6 shows an embodiment of the cartridge unit 420.

[0029] FIG. 7 shows an exploded view of the cartridge unit of FIG. 6.

[0030] FIG. 8 shows a view of the cartridge body 90.

[0031] FIG. 8a shows a diametric section of the cartridge body 90 shown in FIG. 8.

[0032] FIG. 9 shows a view of a bonnet nut 205.

- [0033] FIG. 10 shows a section through the bonnet nut of FIG. 9.
- [0034] FIG. 11 shows a view of the net of lever assembly 304.
- [0035] FIG. 12 shows a view of the lever assembly of FIG. 11.
- [0036] FIG. 13 shows a view of a stop interceptor.
- [0037] FIG. 14 shows a view of the stem member 104.
- [0038] FIG. 15 shows a view of the moveable disc 102.
- [0039] FIG. 16 shows a view of the stationary disc 112.
- [0040] FIG. 17 shows a planer view of stationary disc 112.
- [0041] FIG. 18 shows the stationary disc inserted in the interior of the lower end 62 of cartridge body 90.
- [0042] FIG. 19 shows a planer view of movable disc 102.
- [0043] FIG. 20 shows an integration of the stationary disc 112 and the movable disc 102 inserted in the lower part 62 of the cartridge body 90 when in closure mode.
- [0044] FIG. 21 shows an integration of the stationary disc 112, the moveable disc 102 inserted in the lower part 62 of the cartridge body 90 when in open mode.
- [0045] FIG. 22 shows a view of the cartridge from opening 82 in the closure mode.
- [0046] FIG. 23 shows a section through the cartridge of FIG. 22.
- [0047] FIG. 24 shows a view of the cartridge of FIG. 22 rotated counterclockwise by 90 degrees.
- [0048] FIG. 25 shows a view of a section of the cartridge of FIG. 24 in a plane perpendicular to the plane of FIG. 23.
- [0049] FIG. 26 shows the stationary disc 112, orientation of the moveable disc 102 and a configuration template 122 for receiving stationary disc 112 and moveable disc 102 when the system is in closure mode.
- [0050] FIG. 27 shows an integration of the stationary disc 112, the moveable disc 102, in the configuration template when the cartridge is in closure mode.
- [0051] FIG. 28 shows the stationary disc 112, orientation of the moveable disc 102 and a configurative template 122 for receiving stationary disc 112, moveable disc 102 when the system is in open mode.
- [0052] FIG. 29 shows an integration of the stationary disc 112, the moveable disc 102 in the configuration template 122 when the cartridge is in open mode and maximum rate of flow is enabled through the faucet spout 10.
- [0053] FIG. 30 shows an integration of the stationary disc 112, the moveable disc 102 in the configurative template 122 when the cartridge is in open mode and the lever 20 has turned from its at rest position through its maximum predetermined angle of rotation thereby enabling in reducing rate of flow through spout 10 of the faucet by partially blocking openings 114.
- [0054] FIG. 31 shows a configuration in closure mode of the cartridge of the orientation of lever 20, the stationary disc 112 and the moveable disc 102.
- [0055] FIG. 32 shows a configuration of lever 20, the stationary disc 112 and the moveable disc 102 when the cartridge is in open mode and maximum rate of flow is enabled through spout 10. Lever 20 does not abut stop 61.
- [0056] FIG. 33 shows a configuration of lever 20, the stationary disc 112 and the moveable disc 102 when the cartridge is in open mode and the lever 20 has turned its maximum predetermined angle thereby reducing flow. Lever 20 abuts stop 61.
- [0057] FIG. 34 shows a view of FIG. 31 for a left-handed user.
- [0058] FIG. 35 shows a view of FIG. 32 for a left-handed user.
- [0059] FIG. 36 shows a view of FIG. 33 for a left-handed user.
- [0060] FIG. 37 shows a configuration of the cartridge unit 604 in closed mode. Crank unit 410 and lever unit 304 not shown.
- [0061] FIG. 38 shows a configuration of the cartridge unit 604 in open mode. Crank unit 410 and lever unit 304 not shown.
- [0062] FIG. 39 shows an embodiment of a crank unit 410.
- [0063] FIG. 40 shows an embodiment of a cartridge unit 420.
- [0064] FIG. 41 shows a section view of a detergent activation unit in a vertical plane of symmetry for an embodiment of the detergent activation unit 430.
- [0065] FIG. 42 shows a section view of a valve for regulating detergent fluid from a fluid reservoir to a detergent activation unit 440.
- [0066] FIG. 43 shows a section of an embodiment of a detergent reservoir unit 450.
- [0067] FIG. 44 shows a configuration of the assembling of crank unit 410 and cartridge unit 420, with a section view of the detergent activation unit 430, a section view of the valve regulating unit 440 and a section view of the detergent reservoir unit 450.
- [0068] FIG. 45 shows labeled parts of crank unit 410.
- [0069] FIG. 46 shows the positioning of crank unit 410 on cartridge unit 420. Bonnet nut 205 is not shown.
- [0070] FIG. 47 shows the positioning of the crank unit 410 and lever unit 305 on cartridge unit 420. Bonnet nut 205 is not shown.
- [0071] FIG. 48 shows an external view of an embodiment of the detergent activation unit 498.
- [0072] FIG. 48a shows a view of region 466 for FIG. 48. FIG. 48b shows a view of region 468 for FIG. 48.
- [0073] FIG. 49 shows the labeled parts of the encasement unit 498 for the detergent activation unit in a vertical plane of symmetry for the internal configuration of encasement unit 498.
- [0074] FIG. 50a shows a section view through line 560 in a vertical plane at right angles to the cylindrical height of the encasement unit 498 shown in FIG. 49.
- [0075] FIG. 50b shows a section view in a vertical plane through line 510 at right angles to the cylindrical height of encasement unit 498.
- [0076] FIG. 50c shows a section view in a vertical plane through line 520 at right angles to the cylindrical height of the encasement unit 498.
- [0077] FIG. 50d shows a section view in a vertical plane through line 530 at right angles to the cylindrical height of the encasement unit 498.
- [0078] FIG. 51 shows the labeled parts of detergent activation unit 430 in a vertical plane of symmetry.
- [0079] FIG. 52 shows the labeled parts of detergent activation unit 430 in a horizontal plane of symmetry.
- [0080] FIG. 53 shows a sectional view embodiment of a valve dispensing unit 484 in a diametric plane with 471, 474 and 482 with spring 483 is not shown in said diametric plane.
- [0081] FIG. 54 shows a section view of an embodiment end cap member 475 of the valve dispensing unit 430.

[0082] FIG. 55 shows an embodiment of the spring member for the valve dispensing unit 483.

[0083] FIG. 56 shows a section view of an embodiment of a member 470 for the valve dispensing unit 484.

[0084] FIG. 57 shows a section view in a plane symmetric to the cylindrically disposed height of an embodiment of the piston unit 449 for the detergent activation unit.

[0085] FIG. 58 shows a view through line 500 of FIG. 56 in a plane at right angles to the cylindrically disposed height of member 485.

[0086] FIG. 59 shows a view from the left of piston unit 449 of FIG. 57.

[0087] FIG. 60 shows a section view through line 502 in a plane at right angles to the cylindrically disposed height of piston unit 449 for FIG. 57.

[0088] FIG. 61 shows a section view through line 504 in a plane at right angles to the cylindrical disposed height for member 493.

[0089] FIG. 62 shows a section view in a vertical plane of the cylindrically disposed inlet detergent valve unit 440.

[0090] FIG. 63 shows a section view in a vertical plane of encasement 680 for inlet detergent valve unit 440.

[0091] FIG. 64 shows a relation view in a vertical plane for members 681 and 682 for the inlet detergent valve unit.

[0092] FIG. 65 shows a spring component 621 for the inlet detergent valve unit 440.

[0093] FIG. 66 shows a view in a horizontal plane containing the line 700 of FIG. 67 looking up to 681.

[0094] FIG. 67 shows a view in a horizontal plane containing the line 702 of FIG. 67 looking down to 682.

[0095] FIG. 68 shows a view in a vertical plane of symmetry for members 682 and 632 of the inlet valve unit 440.

[0096] FIG. 69 shows a section outline view in a horizontal plane containing line 704 looking down to 682.

[0097] FIG. 70 shows a view in a vertical plane of symmetry for the internal parts of the valve unit of 440.

[0098] FIG. 71 shows a view in a vertical plane of the configuration of the detergent reservoir unit 450.

[0099] FIG. 72 shows a view in a vertical plane of symmetry of a choke point for enabling flow or disabling of detergent from the detergent reservoir.

[0100] FIG. 73 shows a view of a stopper for enabling or disabling flow of detergent through choke point 720.

[0101] FIG. 74 shows a top view of the detergent reservoir unit 450.

[0102] FIG. 75 shows a section in a vertical plane of the stopper 726 inserted into the choke point 720 in the closure mode.

[0103] FIG. 76 shows a section in a vertical plane of the stopper 726 inserted into the choke point 720 in the open mode.

[0104] FIG. 77 shows a vertical section through the detergent unit and the detergent activation unit when assembled and the invention is in the off mode for the position of the lever.

[0105] FIG. 78 shows a section in a horizontal plane through the detergent activation unit when assembled and the invention is in the off mode. The position of the lever with atop view of the crank unit 410 is also shown.

[0106] FIG. 78a shows a phase of configuration of the detergent activation unit in the closed mode for showing a relation between the input of detergent valve 632, the piston unit and the output detergent valve 482.

[0107] FIG. 79 shows a vertical section through the detergent unit and the detergent activation unit when assembled and the invention is in an activation mode for the position of the lever.

[0108] FIG. 80 shows a section in a horizontal plane through the detergent activation unit when assembled and the invention is in the activation mode for the position of the lever with top view of the crank unit 410.

[0109] FIG. 80a shows a phase of configuration of the detergent activation unit in the activation mode for relating a relation between the input of detergent and the output of detergent. The input valve 632 is closed while the output valve 482 is open.

[0110] FIG. 80b shows position of the piston as related to the opening of valve outlet valve 484 and the closing of valve 632.

[0111] FIG. 81 shows a vertical section through the detergent unit and the detergent activation unit when assembled and the invention is in the activation mode for the maximum predetermined rotation of the lever. The piston unit 491 would have retracted.

[0112] FIG. 82 shows a section in a horizontal plane through the detergent activation unit when assembled and the invention is in the activation mode for the position of the lever in a top view.

[0113] FIG. 82a shows a phase of configuration of the detergent activation unit in the activation mode for a relation between the input of detergent and the output of detergent and the closing of the detergent inlet valve and the closing of the detergent outlet valve.

[0114] FIG. 82b depicts a configuration of the detergent activation unit 430 for closure of outlet valve 484 and the opening of inlet valve 632.

[0115] FIG. 83 shows a vertical section in a plane symmetric to the detergent activation unit and the detergent reservoir in relation to the position of the crank unit 410, the cartridge unit 420 and the lever unit 304 in the inactive or closed mode of the invention.

[0116] FIG. 84 shows a configuration of the lever, the moveable disc 102 and the fixed disc 112 for the closed mode for FIG. 88.

[0117] FIG. 85 shows a vertical section in a plane symmetric to the detergent activation unit 430 and the detergent reservoir unit 450 in relation to the position of the crank unit 410, the cartridge unit 420 and the lever unit 304 in the active or open mode of the invention.

[0118] FIG. 86 show a configuration of lever 20, the moveable disc 102 and the fixed disc 112 in the open mode for FIG. 90.

[0119] FIG. 87 shows configuration 820 for mounting and stabilizing the cartridge unit 420 for use with a wash basin.

[0120] FIG. 88a shows a configuration 861 for interfacing with the cartridge unit 420 and enabling water from an inlet to the cartridge valve 420.

[0121] FIG. 88b shows configuration 863 for directing water from an inlet to the interior passage of 861.

[0122] FIG. 88c shows a section through 963 of 861 shown in FIG. 88b.

[0123] FIG. 88d shows a configuration for directing water from 829 of outlet cartridge mount 820 to a conduit connected to spout 10.

[0124] FIG. 88e shows a section of the assembly of the cartridge 420, bonnet nut 205, and inlet interfacing member 861 to the cartridge mounting 820.

[0125] FIG. 89 shows a section of the assembly of the cartridge 420, bonnet nut 205, inlet interfacing member 861, cartridge mounting and stabilizing member 820 and member 865 and inlet member 869.

[0126] FIG. 90 shows a configuration of the mounting block 981 for the invention in relation to a wash basin.

[0127] FIG. 91 shows a configuration of an under-cover plate for the invention when mounted in relation to a wash basin.

[0128] FIG. 92 shows a configuration of an escutcheon for the invention when mounted with a wash basin.

[0129] FIG. 92a shows a configuration of the spout assembly.

[0130] FIG. 93 shows the first step for mounting and securely attaching the invention to its mounting base with rubber washers 834 and nuts 833. The detergent activation unit is also mounted to the mounting base.

[0131] FIG. 93a shows a front view of the first step for mounting and securely attaching the invention to its mounting base with rubber washers 834 and nuts 833.

[0132] FIG. 94 shows the next phase in the assembling the invention with the cartridge unit 420, then the turning stop restrictor 404, then the bonnet nut 205 and then the cam unit 410.

[0133] FIG. 94a the escutcheon 940 is next assembled.

[0134] FIG. 94b shows the assembly of spout unit 975, the pop-up unit and lever unit 304.

[0135] FIG. 95 shows an assembly of the invention without the detergent reservoir unit 450.

[0136] FIG. 96 shows a configuration of the assembly of the invention when mounted with a wash basin and the detergent reservoir unit 450.

[0137] FIG. 97 shows and exploded view of the configuration of the assembly of the invention.

[0138] FIG. 98a shows an image of the configuration 861' for interfacing with the cartridge unit 420 and enabling water from an inlet to the cartridge valve.

[0139] FIG. 98b shows an alternative configuration 863' with compatible with means for connection to 861' interface, a cartridge mount 820', an input conduit to interface 861' and an output from a cartridge mount 820'.

[0140] FIG. 98c shows a section of an alternative configuration 820' for the mounting of the cartridge unit 420, bonnet nut 205, and inlet interfacing member 861'.

[0141] FIG. 99 shows a section of the assembly of the cartridge unit 420, bonnet nut 205, inlet interfacing member 861', cartridge mounting and stabilizing member 820' and member 863'.

[0142] FIG. 100 shows an alternative configurative 104' for the stem unit of the invention.

[0143] FIG. 101 shows an alternative configuration 89' for the stop interception.

[0144] FIG. 102 shows the cartridge unit assembled with alternative stop 89 and alternative stem 104'.

[0145] FIG. 103 shows an alternative configuration for the lever handle unit for a rectangular cross-section inlet for the cartridge stem.

[0146] FIG. 104 shows a section of the bonnet nut with a circular cross-section for stabilizing the alternative cartridge system.

[0147] FIG. 105 shows a view of the bonnet nut 205 with a circular cross-section 205 for stabilizing the alternative cartridge system.

[0148] FIG. 106 shows a view of the crank unit 810 with a rectangular cross-section for assembly with the stem alternative 104'.

[0149] FIG. 107 shows an alternative configuration for the cartridge unit assembled with stop 89 and crank alternative 810.

[0150] FIG. 108 shows a cam alternative embodiment 410" for activating a detergent activation unit.

[0151] FIG. 109 shows said cam alternative 410" assembled with the cartridge valve unit 420.

[0152] FIG. 110 shows the cartridge valve unit assembled with alternative cam embodiment 410" and lever 20.

DETAILED DESCRIPTION

[0153] Referring to FIG. 1, what is shown is configuration of an embodiment of the invention as seen in a faucet mount in relation to a wash basin.

[0154] Referring to FIG. 2 what is shown a geometric positioning relation of the base 12 of a faucet spout 10, and distance 14, labeled a of said base 12 of said faucet spout 10 from said base 12 of spout 10 to a base 16 of a lever handle 20 of said faucet where said lever handle 20 is enabled with means for rotational motion when said lever handle 20 is attached to cartridge assembly 420. Said lever handle 20 rotation being enabled by connection to the upper portion of a spindle of a faucet cartridge valve. Said lever 20 is preferably configured with an upper flat surface region 201 and a lower curved surface region 202 to better to facilitate disbursement of liquid on said upper flat surface region 201 and adhesion of liquid to said lower curved surface region 202 thus substantially facilitating washing of total surface area of said lever 20 when said lever 20 is disposed to intercept line of flow 777 from spout portion 8. The members detergent reservoir 450 and spout escutcheon 977 are not shown in FIG. 2. Said lever 20 is preferably made from brass or stainless steel.

[0155] Referring to FIG. 3, what is shown is a view from the right of FIG. 2. Line 777 shows a direction of vertical flow of fluid from a faucet spout portion 8 of said spout 10 on said faucet mount. The horizontal distance 24, labeled b, between line 444 and line 777 and in the plane containing line 444 and line 777 is such that the vertical line 555 projects between line 444 and line 777 and in the plane containing line 444 and line 777 and where line 555 is the vertical line of axis 555 of said cartridge unit 420.

[0156] Referring to FIG. 4 and FIG. 2 geometric length 26 labeled c of said handle lever 20 can be related to the said geometric distance a and said geometric distance b by the inequality $\sqrt{a^2+b^2} \leq c$. When said relation $\sqrt{a^2+b^2} \leq c$ exists in said faucet said lever handle 20 can be enabled to intersect said line of fluid flow 777 from said spout portion 8 of spout 10 of said faucet; thus said lever portion 20 can be exposed to interact with said fluid flowing from said spout portion 8 of said spout 10.

[0157] Referring to FIG. 5, what is shown is a geometric configuration for a single lever faucet unit.

[0158] Referring to FIG. 6, what is shown is an embodiment of a cartridge unit 420. Referring to FIG. 7 what is shown are the exploded components of said cartridge unit 420. As shown in FIG. 6 cartridge 420, includes a body 90, said body 90 has an exterior surface 64 and an interior passage 68. FIG. 8 shows a configuration of said body 90 while FIG. 8a shows the configuration of a cross-section of

said body 90. As shown in FIG. 8 the exterior surface is preferably generally cylindrical in shape so as to be easily situated in a flow conduit for fluid and thus enable the action of an inline valve between the conduit and the outlet of a faucet. The body may be formed of a variety of materials such as metal, metal alloys or other utility compatible economical nonmetallic composite or thermoplastic.

[0159] The interior passage 68 of the body 90 defines a longitudinally extending passage 68 through the body as shown in FIG. 8a. Passage 68 preferably extends from an inlet opening 70, at the bottom of the body to a lateral outlet opening 82 at the top of the body. Passage 68 preferably has a cylindrical configuration conducive to use of rotating parts therein and can also have one or more sections or ones to accommodate the size and shape of various parts therein. As shown the passage has an upper portion 76 which is preferably of a narrower cross-sectional diameter as shown in FIG. 8a to correspond to the dimensions of the region 57 of said stem 104, and a lower wider cross-sectional diameter section 78 that is preferably wider to accommodate other parts of the valve such as a disk, sealing rings and rotatable base portion of said stem 104.

[0160] The body further includes at least one side opening 82 extending transversely through the body from the interior passage 68 to the exterior surface of the body 90. The side opening 82 is preferably located to be in keeping with current faucet design in that water or other fluid flowing upwards through the inlet end 70 of the body into passage 68 would encounter the stationary disk and pass through the opening therein when in the open valve mode until it entered the receiving area of the moveable disk and could then flow out the side opening 82 in the body to continue on to feed the faucet body and outlet of the faucet.

[0161] Thus, one or more side opening 82 for this embodiment should preferably be positioned to accommodate that flow out said side openings 82 in an unimpeded manner within the design. The opening is preferably rectangular for fluid flow characteristics but can have other configurations. As shown two such side openings 82 are shown which is compatible with the number of openings and receiving areas in the disk cartridge.

[0162] The openings 82 is preferably rectangular for fluid flow characteristics but can have other configurations. As shown two such side openings 82 are shown which is compatible with the number of openings and receiving areas in the disk cartridge.

[0163] FIG. 9 shows a configuration of body stop interceptor member 205 while FIG. 10 shows a cross-section of said member 205.

[0164] Referring to FIG. 8, said body 90 has an upper substantially flat surface 130 enabled with means for limiting the rotation of handle 20 and thus the rotation of stem 104. Said means for limiting the rotation of stem 104 is enabled by raised feature 60 and raised feature 61 where said feature 60 and said feature 61 are designed to limit the turning of said stem 104 by said lever 20 to a maximum predetermined angle of rotation; such limiting is enabled when said stop 60 and said stop 61 are rigidly connected to said upper surface 130 of said cartridge body 90. As will be shown said maximum predetermined angle of turn limits the range of motion of a moveable disc 102 against the openings of a fixed disc 112 and thus enables desired coordination of the various flow openings within a particular predetermined rotational range. The embodiment of such a moveable disc

is shown in FIG. 15 while the embodiment of such a fixed disc is shown in FIG. 16. Said stop feature 60 and said stop feature 61 are enabled on said upper surface 130 and arranged such that with the centre of a circle lying on the axis of rotation of stem portion 57, said stop features location 60 and 61 may characterize end points on separate radii where said radii originate from a common point lying in the plane of surface 130 on the axis of rotation of stem portion 57.

[0165] Referring to FIG. 10, interior flange 47 of bonnet nut 205 shown in FIG. 9 contacts outer region of body surface portion 130. Access region opening 45 of said bonnet nut 205 mates with and surrounds region 59 of stem 104; however, said access opening 45 of said bonnet nut 205 is enabled to rotate substantially freely around region 59 of stem 104. Said bonnet nut 205 fully accommodates passive stop 60, passive stop 61 and said active stop 404 such that a measurable distance exists between the interior region of 205, passive stop 60, passive stop 61 and active stop 404 thus enabling stability of stem 104 when said active stop 404 is mated to region 56 of said stem 104. Member 65 of active stop 404 interact with static stop 60 and static stop 61 to substantially enable the maximum predetermined angle of rotation of stem 104. The bonnet nut 205 functions to ensure stability of the stem 104 and by extension, the stable function of the cartridge unit 604.

[0166] Referring to FIG. 14, what is shown is cartridge unit member 104 generally called the stem of a ceramic faucet unit. Said stem 104 includes a reduced cylindrical diameter region 57 of said stem member 104. The upper region 58 of said region 57 is contiguous with a lower region 52 of said region 57. Said region 58 is configured to be in rotatable relationship with lever handle unit 304 of FIG. 12 while said region 52 will be shown to be configured to be in rotatable relation with faucet system member 410 shown in FIG. 39.

[0167] Contiguous to region 52 is region 59 of said region 57. Said region 59 is configured to be substantially frictionless and configured to move substantially frictionless against the interior 45 of component member 205 shown in FIG. 10.

[0168] Contiguous to region 59 of said region 57 is region 56 of said region 57. Said region 56 of FIG. 14 is configured to be in rotatable relation with interior 34 of 404 depicted in FIG. 13.

[0169] Referring to FIG. 13, faucet unit member 404 when assembled with region 56 of upper stem portion 57 functions to interact with body stops 60 and 61. Said member 65 of member 404 will be shown to interact with body stop 61 and body stop 60 and also will be shown to enable a predetermined angular rotation of region 57 and consequently, ensure rotation of a moveable disc 102 by said stem 104.

[0170] Referring to FIG. 12 portion 58 of stem 104 is enables rotation of said stem 104 when said stem 104 is in locking relationship with lever unit 304; such locking means between exterior region 58 of region 57 of 104 and interior 34 of lever unit 304 may be enabled by a grooved locking between region 58 of portion 57 and the interior wall of circular slots 34 of said 304. The interior wall 34 of 410 is also grooved for locking means with region 52 of region 57 of said stem 104 while the interior 34 of 404 is grooved for locking means with region 55 of region 57 of said stem 104. Region 59 of region 57 is substantially smooth and enables rotation of the interior surface 45 of 205 to substantially frictionless motion on the said surface 59 of region 57. FIG.

9 shows a configuration of member 205 while FIG. 10 shows a cross-section of said member 205. It will be shown that such locking may be enabled by rectangular slots.

[0171] As shown in FIG. 14 region 57 is part of the stem 104 of said cartridge unit 604. Said region 57 has an upper portion 58 configured to fit in rotatable relationship with the interior surface 34 of lever assembly 304 shown in FIG. 12, and with the lower portion 66, configured to be accommodated within the body 90 when said stem 104 is situated within the passage 68 of said body 90.

[0172] As shown in FIG. 14 region 57 is part of stem 104 of said cartridge unit 604. Said region 57 has an upper portion 52 contiguous to upper region 58 configured to fit in rotatable relationship with the interior surface 34 of crank assembly 410 shown in FIG. 39, and with the lower portion 66, configured to be accommodated within of the body 90 when said stem 104 is situated within the passage 68 of the body 90.

[0173] As shown in FIG. 14 the stem 104 has a region 59 configured to rotate independently and substantially frictionless within member 45 of bonnet nut 205 shown in FIG. 9 and FIG. 10 and with the lower portion 66, configured to be accommodated within the interior of the body 90 when said stem 104 is situated within the passage 68 of said body 90. Said region 59 is contiguous to member region 52.

[0174] As shown in FIG. 14 said region 57 of stem 104 has portion 56 configured to fit in rotatable relationship with the interior surface 34 of stop assembly 404 shown in FIG. 13, and with the lower portion 66, configured to be accommodated within the interior of the body 90 when said stem 104 is situated within the passage 68 of the body 90. Said region 56 is contiguous with region 59.

[0175] Referring to FIG. 14 the stem 104 of cartridge 604 has a lower portion 66 which is configured to fit in rotatable relationship with the interior passage 68 of the body 90 when said stem 104 is situated within the said passage 68 of said body 90. As shown in FIG. 14 the upper portion 55 of said stem 104 is enabled with groove 85 for receiving at least one sealing ring 72. Referring to FIG. 14, when said stem 104 is integrated within passage 68 of the body 90 said sealing ring 72 functions to prevent leakage around the stem in said region 55. As shown in FIG. 14 lower portion 66 of said stem 104 is enabled with a groove 87 at least one sealing ring 74. Referring to FIG. 14 when said stem 104 is integrated within passage 68 of the of said body 90 said sealing ring 74 functions to prevent leakage between the interior surface of the said body 90 and the lower end portion 66 of said stem 104. Said sealing ring 74 sits in inclusion 87 of said stem 104.

[0176] The upper end region 57 of the stem 104 may preferably have a threaded or other snap-fit, slide in, friction fit or gripping surface for attaching the cartridge to a faucet handle so as to be rotatable and function with the handle 20. The lower end portion 66 of the stem 104 has a bottom surface 96 to which is attached a feature 98. The upper end portion 58 of the said region 57 as shown in FIG. 6 is configured to be fitted within a faucet handle assembly 304 shown in FIG. 12. The stem 104 may be formed of a one-piece construction from stainless steel, brass or zinc alloy or copper alloy or other safe metal plastic.

[0177] The bottom surface 96 of said stem 104 preferably has rectangular prismatic feature 98, extending from said surface 96. Said rectangular prismatic feature 98 when within tolerance is accommodated within rectangular pris-

matic slot 100 of moveable disc 102 will be shown to allow for interaction with said moveable disc 102 so that when said upper portion of spindle 58 region 57 of said stem 104 is fitted within a faucet handle assembly 304 and lever 20 is turned a predetermined angle, it will then in turn rotate the said moveable disc 102 substantially that of said predetermined angle of said lever assembly 304 and consequently activate the open/closure feature of the cartridge valve. FIG. 12 shows a configuration of a faucet handle assembly 304. FIG. 15 shows a configuration of a moveable disc.

[0178] Referring to FIG. 14, a region defined by 59 of region 57 moves freely substantially frictionless within reduced diameter region 45 of bonnet nut 205. Said bonnet nut functions to substantially ensure stability of the cartridge unit.

[0179] Referring to FIG. 15 what is shown is a configuration of the moveable disc 102. Said moveable disc 102 is enabled with an inclusion 110, said inclusion 110 is contained by parabolic arc 91, and mirrored parabolic arc 94, parabolic arc 92 and mirrored parabolic arc 95, line segment 96 and mirrored line segment 97. Said inclusion is further defined by line segments 95 and line segment 99. Said inclusion 110 is mirrored on an opposite surface of said moveable disc 102. Said moveable disc 102 is further enabled with a preferably prismatic inclusion 100 where said inclusion 100 functions to tolerably accept extended tab 98 of the stem 57 thus enabling turning of moveable disc 102 against fixed disc 112 when said unit 57 is activated to rotate by activation of lever 20 for its utility and disc 102 enables fluid through 114 of fixed disc 112 into harbor 110 of said moveable disc 102.

[0180] Additionally, the upper surface of said moveable disc 102 is disposed to be described as a section of a cross-section of a cylinder defined by two parallel chords shaped to fit within a generally cylindrical portion of the passage 68 in the lower portion 78 of said passage 68. Said diametric inclusions 110 in the lower portion of moveable disc 102 function as fluid receiving harbors is the said moveable disc 102.

[0181] The movable disk 102 included within the body in passage 68 is enabled with means on its upper surface 104 configured to contact and interact with the bottom surface 96 of the lower end portion of the stem 66 in a rotatable relation. As shown, upper surface 104 is preferably configured with a slot 100 for receiving a tab 98 therein. The disk upper surface may preferably be defined as a portion of the cross-section of a solid cylinder defined by symmetrically parallel chords shaped to fit within a preferred generally cylindrical portion of the passage 68 in the lower portion 78 in the passage 68. There are preferably two side wall portions defining preferably two fluid receiving areas 110 within the moveable disk 102. The receiving area 100 is configured so that when the disk 102 is rotated to align the receiving area 100 with the side opening 82 in the body 68 it will allow fluid flow out of the side opening of the body for feeding the faucet to which is installed with fluid entering the cartridge from a conduit through the open inlet 70 of the cartridge.

[0182] Referring to FIG. 16 what is shown is a stationary disk 112. Said disk 112 is fixed in position for interacting with moveable disc 102. Said fixed disc 112 is preferably provided with openings 114 extending through the stationary disk 112 for enabling passage of fluid flow from an inlet end 70 of the body 90 then into the passage 68, and through

the stationary disk and then into the receiving area 110 of the moveable disk, and then out the side opening 82 of the body when the cartridge is in the open position mode. The stationary disk 112 is preferably enabled with lubricant containing pits 115; said lubricant in lubricant containing pits function to reduce friction between moveable disc 102 and stationary disc 112 thus further prolonging functional integrity of said moveable disc 102 and said stationary disc 112. Said fixed disc is further provided with diametric tabs 118; said tabs 118 enables the said disc 112 to be fixed in recesses in the interior passage 68 of the body cartridge 90.

[0183] Referring to FIG. 17 what is shown is another view of the stationary disc 112. Referring to FIG. 18, what is shown is a configuration of said fixed disc 112 positioned with diametric tabs 118 recessed in said body 90 of the cartridge unit. Referring to FIG. 19 what is shown is a view of the moveable disc 102. Referring to FIG. 20 what is shown is a relationship between said moveable disc 102 and said stationary disc 112 when cartridge valve unit 604 is in closure mode thus substantially resisting the flow of liquid from conduit 70 through passage 68 and to side openings 82. Referring to FIG. 21 what is shown is a relationship between said moveable disc 102 and said stationary disc 112 when said cartridge valve unit 420 is the open mode thus enabling the flow of liquid from conduit 70, through the passage 68 and to side openings 82. As will be shown said disabling and enabling of the flow of liquid through the valve cartridge unit is enabled when a lever handle 20, when fitted to portion 58 of the region 57 of the stem 104 shown in FIG. 14 can be turned to reduce and shut off flow. The handle 20 thus initiates the closing of the valve from the open mode of said valves and the opening of the valves from a closure mode of said valves when said handle 20 is initiated to turn.

[0184] Referring to FIG. 22 what is shown is a view from the side opening 82 of said cartridge unit 420 when said cartridge unit 420 is in closure mode. FIG. 23 shows a diametric cross-section of the view of FIG. 22 and here said diametric cross-section lies in a plane of symmetry across the most substantial dimensions of preferably rectangular prismatic slot 100 of said moveable disc 102.

[0185] Referring to FIG. 24, what is shown is a view of said cartridge 420 of FIG. 22 rotated through 90 degrees. FIG. 25 shows a diametric cross-section of the view of FIG. 24 where said diametric cross-section lies in a plane of symmetry across the lesser dimensions of preferably rectangular prismatic slot 100 of said movable disc 102.

[0186] Referring to a cartridge configuration say FIG. 23 for clarity; for the washing of the lever 20 the turn of stem 104 when lever handle 20 is attached to said stem portion 58 of stem 104 enables opening and closing of valves 102 and 112 and thus enables utility of the invention. The washing of the lever 20 of the invention is enabled when said lever arm 20 is attached to stem region 58 of said stem 104 and said lever 20 is initiated to rotate through a maximum predetermined angle that is greater than the angle turned to enable said lever 20 to intercept the said line of flow 777 from spout 10. The initial maximum predetermined turn of the stem 104 by lever 20 from an off mode of the cartridge 420 will be shown to reduce flow rate through spout 10 to enable spreading of detergent over ones hand with the option for washing of hands; a subsequent counter turn of said lever 20 through a predetermined angle enables increase in the flow rate through spout 10 for washing of hands using a greater flow rate; a subsequent counter turn through a predetermined

angle enables said lever 20 to intercept line of flow 22 from spout 10 and also increases flow rate through spout 10 to its predesigned maximum thus enabling the choice of washing of said lever 20 and further washing of hands. A subsequent counter turn of said lever 20 through a predetermined angle enable repositioning of lever arm 20 back to a rest positioning in which flow of liquid is disabled. The reduction in flow rate of solvent through spout 10 thus enables significant conservation in the use of water as a solvent and the washing of said lever 20 significantly reduces the spread of pathogens thus enhancing health and safety.

[0187] Referring to FIG. 16 said stationary disc 112 is enabled with preferably diametric paired extension 118 for locking said stationary disc 112 within the interior passage 68 of the cartridge body 90; thus, enabling reliability and stability of said cartridge unit 420. As shown in FIG. 18, said means for locking said stationary disc 112 may be achieved when inclusion is designed within the interior passage 68 of said body 90 and were said inclusions in the passage 68 of said body 90 are dimensioned to accommodate and fit said diametric extensions 118 of said stationary disc 112. Said stationary disc 112 interfaces with the moveable disc 102 located in the passage 68 of the cartridge body 90 and with conduit 70 where preferably openings 114 in said stationary disc 112 accommodate flow to the receiving are 110 of moveable disc 102 and then through side openings 82 to the exterior of body 90 of cartridge unit 420.

[0188] Referring to FIG. 25 said conduit 70 is external to said interior passage 68, thus, to ensure reliability of the cartridge unit an elastomeric seal 88 is introduced to interface said stationary disc 112 and said conduit 70 in region 132 of the cartridge body 90. Said sealing ring 88 included within component 132 functions to apply pressure between 132 stationary valve member 112, and moveable valve member 102. The pressure exerted by said scaling ring 88 ensures optimal functioning of the cartridge for the flow of liquid. The stationary disc 112 is also enabled with a pit like lubricant containing feature 115; lubricant in said pit like feature 115 function to for ensure substantially frictionless movement of said moveable disc 102 over the surface of fixed disc 112 and also to prolong the functional integrity of the cartridge unit 420.

[0189] Referring to FIG. 26; what is shown is an explosion of the order of contiguous assembly of the position of tabs 118, the orientation of the moveable disk 102 and the orientation stationary disk 112 and a cross-section through the lower part 132 of the said cartridge body 90 when the faucet system shown in FIG. 1 is in the off mode but disposed to enable flow of liquid to said faucet when activated. Referring to FIG. 27 in contrast to FIG. 26, what is shown is an integration and relative contiguous orientation of said moveable disk 102 and said stationary disk 112 in the passage 68 of the lower part 132 of said body 90 when the faucet system of FIG. 1 is in the off mode but disposed to enable flow of liquid to said faucet when activated.

[0190] Referring to 28; what is shown is an explosion of the order of contiguous assembly of the position of tabs 118, the orientation of the moveable disk 102, the orientation of stationary disk 112 and a cross-section through the lower part 132 of the said cartridge body 90 when the faucet system shown in FIG. 1 is in the activated mode but disposed to enable substantial flow of liquid through spout 10 when said lever 20 does not intercept the line of flow 777 from spout 10 to a wash basin. Referring to FIG. 29 in contrast to

FIG. 28, what is shown is an integration and relative contiguous orientation of said moveable disk 102 and said stationary disk 112 in the passage 68 of the lower part 132 of said body 90 when the faucet system of FIG. 1 is in the activated mode but disposed to enable substantial flow of liquid through spout 10.

[0191] Referring to 30, what is shown is an integration and relative contiguous orientation of said moveable disk 102 and said stationary disk 112 in the passage 68 of the lower part 132 of said body 90 when the faucet system of FIG. 1 is in the activated mode but disposed to reduce flow of liquid through spout 10 where said reduction in flow rate of liquid conserves water as a solvent when hands are being soaped.

[0192] Referring to FIG. 31; what is shown is the order of contiguous assembly and relative orientation of the handle 20, the orientation of the moveable disc 102 and the orientation of the stationary disk 112 when faucet system is in the off mode and liquid flow is disabled through spout 10 of faucet assembly for a right hand side position of faucet cartridge 420.

[0193] Referring to FIG. 32, moveable disc 102 is shown with a line of symmetry 1020 for said moveable disc 102; now referring to 1020 of FIG. 31 and consistent with the properties of a rigid body, said 1020 characterizes initial orientation of said line of axis of symmetry 1020 for 102 of FIG. 31 and transformation of said axis of symmetry 1020 in 102 in FIG. 32 subsequent to lever 20 executing its predetermined angle of rotation; thus indicating a sense and angle of rotation of moveable disc 102 from the inactive mode to the utility mode of configuration where said lever arm 20 is not disposed to intercept line of flow 22 from portion 8 of spout 10.

[0194] Referring to FIG. 33 the predetermined maximum turn of moveable disc 102 is shown with a line of symmetry 1020 for said moveable disc 102; now referring to 1020 of FIG. 32 and consistent with the properties of a rigid body, said 1020 characterizes initial orientation of said line of axis of symmetry 1020 for 102 of FIG. 32 and transformation of said axis of symmetry 1020 in 102 in FIG. 33 subsequent to lever 20 executing its predetermined maximum angle of rotation; thus when the cartridge system 604 exists in this mode the flow rate of solvent say water is significantly reduced for enabling soaping of hands and thus as mentioned enabling economic utilization of water and by extension economic, environmental and health conservation because the lever is now washed by hands that have been washed before the cartridge system is returned to its inactive mode by the turning of said lever 20 and thus also mitigating the spread of pathogens. The washing of lever 20 is enabled when said lever 20 is counter turned to intercept the line of flow 22 of liquid from spout 10 thus said lever 20 is disabled from the line of flow of solvent from spout 10 when hands are being washed.

[0195] Referring to FIG. 34, FIG. 35 and FIG. 36 what is shown is a configuration of FIG. 31 FIG. 32 and FIG. 33 for an embodiment of the invention in which the said cartridge valve 420 is configured for activation by turning a lever counterclockwise for said systems activation. Such a system is a configuration in which the cartridge unit is left side of the base of a faucet basin and more suited for a left-handed person.

[0196] Referring to FIG. 37, what is shown is an embodiment of a right-hand side cartridge valve 420 in the inactive

mode were said member 65 attached to said stem 104 would abut stop member 60 to effect closing of openings 114 of stationary disc 112.

[0197] Referring to FIG. 38 what is shown is an embodiment of a right hand side cartridge valve 420 in the active mode and where rotation of the stem 104 through a maximum predetermined angle from the off configuration of the cartridge 420 would have been limited by stop member 61 when said member 65 of said stem 104 intercepts body stop member 61.

[0198] Referring to FIG. 39, what is shown is an embodiment of the crank attachment component unit 410. FIG. 40 shows an embodiment of the cartridge unit 420, bonnet nut 205 is not shown. FIG. 41 shows a substantially symmetric section in a vertical plane of an embodiment of a detergent activation unit 430. FIG. 42 shows a substantially symmetric section in a vertical plane of an embodiment of a valve component unit 440. FIG. 43 shows a substantially symmetric section in a vertical plane of an embodiment of a detergent reservoir unit 450. FIG. 44 shows an integration of components units 410, 420, 430, 440 and 450.

[0199] Referring to FIG. 45, crank component unit 410 shows a crank region 460 with a connecting rod 496. Said crank unit 410 has a cylindrically inner grooved attachment region 462. Said inner grooved attachment 462 locks to region 52 of the upper stem portion 57 of stem 104 in a complete unit for the invention such that rotation of stem 104 substantially simultaneously rotates said crank component unit 410. Said crank unit component 410 is preferably made from non-corrosive metal such as copper, brass in order to ensure maximum and prolonged functional reliability.

[0200] Referring to FIG. 46, what is shown is an integration of said crank attachment component unit 410 to portion 52 of upper portion 57 for the said stem unit 104 of cartridge unit 420. Said crank component unit 410, said lever 20 and said upper portion 57 constitute a rigid body thus consistent with the properties of a rigid body said crank 460 will rotate consistent with the sense of clockwise or counterclockwise rotation of said lever 20 and additionally said crank 460 rotation occurs with substantially relative invariance in angular configuration relative to said lever 20. The vertex of the angle characterizing said angular separation lies on a common axis of rotation of said stem 104 and said lever 20 consequently, enabling substantial invariance in a predetermined said angular separation of said crank component 460 and said lever 20 when said crank 460 and said lever 20 are attached to said stem 104. Said crank 460 is thus substantially constrained to move in a plane. Additionally, said crank 460 and said lever 20 are disposed to be separately mated to said upper stem portion 57 by groove fit thus enabling ease of manipulation in positional variation in said angle of separation between said lever 20 and said crank 460.

[0201] Additionally, a connecting rod 496, freely hinged to said crank 460 enables translation of a piston 491 when said connecting rod 496 is attached to a piston 491 thus enabling said piston 491 to activate opening and closing of valves to enable dispensing of liquid detergent from a detergent activation unit 430. Said another way, the connecting rod 496 and the crank 460 form a two-bar linkage. Furthermore, the function of said crank 460 and said connecting rod 496 is enabled on multiple angular position on said stem 104 and so enables convenient positioning of the

detergent activation unit for initiating dispensing of liquid detergent by the action of said crank **460**. A detergent dispensing system can therefore be conveniently positioned on a faucet mounting base for interacting within an annulus for the crank unit **410**. FIG. **46** functions to show a positioning of the crank unit **410** relative to the lever assembly **304**; in an assembly of the invention the positioning of the bonnet nut **205** would precede the positioning of the crank unit **410** to be followed by the positioning of the lever unit **305**.

[0202] Referring to FIG. **47**, what is shown is an integration of crank unit component **410** and lever unit **304** to said upper portion **57** of stem unit **104** of cartridge unit **420**; thus as stated said integration of crank unit **410** and lever **20** to said upper portion **57** substantially constitutes a rigid body, consequently initiation of rotation of lever **20** initiates rotation of stem **104** and thus rotation of crank unit **410**. FIG. **47** functions to show the positioning of the crank unit **410** relative to the lever assembly **304**.

[0203] Referring to FIG. **48**, component **498** shows an embodiment of an encasement unit or encasement member **498** for coupling with a detergent activation unit **430**. Referring to component **498**, said component **498** is disposed to be characterized as an open cylinder **464** contiguous with diametric prismatic flanges **465**. Said prismatic flanges **465** are enabled with access openings **497**; said openings **497** function as a means for attaching encasement **498** firmly to a wash basin faucet mount. Said cylindrical component **464** is further enabled with a spirally grooved opening **487**; said spirally grooved opening **487** will be shown to enable the attachment of a liquid detergent reservoir member **450**. Additionally, said cylindrically disposed region **464** is enabled with means for attaching a conduit **478**; said conduit functions to direct liquid detergent from the interior of said cylindrically disposed component **464**. Said cylindrically disposed region is further characterized by region **466** and a region **468**. Said detergent activation encasement **498** is preferably configured from Bakelite, melamine or nylon.

[0204] Referring to FIG. **48a**, what is depicted is region **468** of said cylindrically disposed member **464**. Said cylindrically disposed region **468** is internally spirally grooved and will be shown to be compatible with a closed end cap **475**.

[0205] Referring to FIG. **48b**, what is depicted is region **466** of said cylindrically disposed member **464**. Said region **466** of cylindrically disposed region **464** is enabled with guide channels **492** where said guide channels will be shown to restrict a piston **491** substantially to translation motion.

[0206] Referring to FIG. **49** what is shown is an embodiment of a vertical planar cross-section of said cylindrically disposed component **464** for encasement **498** where said planar cross-section may contain the cylindrical axis of that said cylindrically disposed **464** and that axis contains the center of a circular cross-section of that said cylinder **464**. Said cylindrically disposed **464** is also enabled with grooves for containing sealing rings **480**, **481** **489** and **490**. In an assembly of the invention said sealing rings **481**, **480**, **489** and **490** function to substantially resist leakage of detergent fluid from internal region **487** of said cylindrically disposed **464**.

[0207] Referring to FIG. **49**, said cylindrically disposed component **464** has its internal region modified for insertion, translation, stability and optimal function of a piston unit component **449**.

[0208] FIG. **50a** shows a conceptual cross-section of said **464** in a plane containing the line **560** and were said plane containing line **560** is perpendicular to the cylindrical axis of said cylindrically disposed **464**.

[0209] FIG. **50b** shows a conceptual cross-section of said **464** in a plane containing the line **510** and were said plane containing line **510** is perpendicular to the cylindrical axis of said cylindrically disposed **464**. Said cross-section defined by FIG. **50b**, in a region of a certain length and on the interior wall of cylindrically disposed **464**, is enabled with detergent fluid outlet conduit attachment **478**. Said outlet attachment is sandwiched between sealing rings **480** and sealing ring **481**. It will be shown that in an assembly of a properly functioning unit of the invention an out-valve unit **482** is excluded from travel to region of a certain length defined by line **510** and line **560** of said **464**.

[0210] FIG. **50c** shows a conceptual cross-section of said **464** in a plane containing the line **520** and were said plane containing line **520** is perpendicular to the cylindrical axis of said cylindrically disposed **464**.

[0211] FIG. **50d** shows a conceptual cross-section of said **464** in a plane containing the line **530** and were said plane containing line **530** is perpendicular to the cylindrical axis of said cylindrically disposed **464**. Diametrically paired slots **492** in a region of a certain length would contain line **530**. It will be shown that said diametric slots function to substantially resist rotation of a piston unit **449** when diametrically paired flanges **495** are enabled on said piston **491** such that said piston **491** is forced by said flanges to translate in said slots **492**. Said diametrically paired slots **492** of cylindrically disposed **464** do not extend to regions of said cylinder **464** were said slots **492** would impinge the function of o-rings **489** and **490**. Sealing rings **489** and **490** are enabled with access within regions defined by line **520** and line **530** of the interior wall of said cylindrically disposed **464**.

[0212] Referring to FIG. **49**, said cylindrically disposed component **464** is enabled with a detergent containment region **487**. In a functioning unit of the invention liquid detergent is enabled to enter said detergent region **487** after detergent fluid in said region **487** has been substantially expelled. The interior wall of region **487** of cylindrically disposed **464** is enabled with a circular flange **477**. It will be shown that said circular flange **477** function to contain an outlet valve unit **482**.

[0213] Referring to FIG. **51**, what is shown is an embodiment of a planar cross-section of the detergent activation unit **430**. Said detergent activation system **430** comprises a stabilizing encasement component **498** which has already been described, an outlet valve component unit **484** shown in FIG. **53** and a piston component unit **449** shown in FIG. **57**. Said piston component **449** is preferably configured from Bakelite, melamine or nylon.

[0214] Referring to FIG. **53**, what is shown is an embodiment of the detergent outlet unit **484**. FIG. **54**, FIG. **55** and FIG. **56**, show an embodiment of the individual members of said detergent outlet **484**. Referring to FIG. **54**, what is shown is an embodiment of an end cap component **475** of said detergent outlet unit **484**. Said component **475** is disposed to be characterized as a solid cylinder portion **471**

contiguous with a an outer spirally grooved cylindrical annulus 474 to enable spiral insertion in compatible grooved region 468 of said cylindrical disposed 464. Said outer spirally grooved component 474 is of a lesser radius than cylindrically disposed component 471. An outer grooved preferably solid cylindrical region 473 of a lesser radius than 474 is contiguous with a circular face member 471 but non-contiguously included by cylindrically shaped annulus 474. Referring to FIG. 55, what is shown is a spring 483 dimensioned to be spiraled within the grooves 472 of 473 thus enabling one end of said spring 483 to be fixed when said 475 is attached in region 468 of cylindrical disposed 464. Components 471, 474 and 473 are disposed to be geometrically concentric. Referring to FIG. 56, component 470 characterizes cylindrically disposed contiguously concentric with a lesser radius cylindrically disposed region 476. Said cylindrically disposed region 476 is spirally grooved to be compatible with spring member 483. Said grooved member 476 functions to stabilize said spring 483 and preferably fully enshrouds said member 485. When said 475, 483 and 470 are integrated in a functioning unit of the invention the sum total of the cylindrical heights of cylinder 473 and the height of cylinder 476 is less than that of the relaxed length of spring 483 thus enabling a compressive length of said spring 483. It will be shown that the compressive length of said spring 483 is predetermined to be greater than the maximum translation of a piston unit 449 when said invention functions according to design. Said outlet valve 484 components may preferably configured from Bakelite, melamine or nylon while the spring unit 483 is preferably made from stainless steel for resistance to the humid environment of a faucet.

[0215] Referring to FIG. 57, what is shown is an embodiment of the piston component unit 449. Said piston component 449 is preferably cylindrically disposed and is enabled with a region 491 where said region 491 is predisposed to move substantially frictionless against the inner wall of said cylindrically disposed 464. FIG. 58 shows a cross-section of said 491 through a line 502 where said line 502 lies in a plane perpendicular to the central axis of cylindrically disposed 464. Said component 449 is further enabled with a beveled upper component 488 where said upper beveled component 488 is predetermined to function as a wedge for initiating motion of a valve 632.

[0216] Referring to FIG. 57, contiguous and concentric to said cylindrically disposed 491 and in region 487 of said cylindrically disposed 464 is component 486. Said component 486 is of a lesser radius in comparison to the radius of 491. Said reduced radius component 486 functions to enable detergent liquid availability in said region 487 and also to initiate dispensing of fluid detergent from said region 487. It will be shown that said reduced radius of 486 in comparison to the radius of member 491 functions as a leading edge of the piston unit 449 for initiating contact with 482 of outlet valve unit 484 to enable detergent liquid dispensing through outlet 478 and for replenishing region 487 of 498 with detergent fluid in said region 487. Contiguous with said component 491 and in region 466 of cylindrically disposed 464 are flange component 493 and flange component 494. Said flanges 493 and 494 are enabled with access slots 454 and 455. Said access slots 454 and 455 enable hinged connection of piston unit 449 to connecting rod 496.

[0217] Referring to FIG. 57, said piston component 449 is diametrically enabled with raised flanges 495. Said diametri-

cally paired flanges are preferably vertically disposed and positioned in diametric access slots 492 in region 466 of said cylinder 464. Said diametrically paired flanges are predetermined to function within the limits of paired access slots 492. Said diametrically paired flanges 495 function to resist rotation in the movement of piston unit 449 and thus substantially restrict the motion of piston unit 449 to translation. Additionally, said diametric flanges 495 enable piston component to stay within the influence of sealing ring 489 and sealing ring 490.

[0218] Referring to FIG. 58, what is shown is a cross-section through line 500 of FIG. 53. Referring to FIG. 59, what is shown is a view of piston component 449 of FIG. 57 from the left. Referring to FIG. 60, what is shown is a cross-section projection of component 491 of FIG. 57 in a plane containing line 502 and where that plane is perpendicular to the height of cylindrically disposed 449. Referring to FIG. 61, what is shown is a cross-section projection of component 491 of FIG. 57 in a plane containing line 503 and where that plane is perpendicular to the height of cylindrically disposed 449.

[0219] Referring to FIG. 52, what is shown is a section view of said detergent activation unit 430 in a horizontal plane where said horizontal plane contains a section view of detergent outlet conduit 478. Access slots 497 enable firm attachment of said detergent activation unit 430 to the mounting base for a faucet such that said detergent activation unit 430 is within influence of the crank 460 and follower 496 in a functioning unit of the invention.

[0220] Referring to FIG. 62, what is shown is an embodiment of a vertical semicircular disposed cross-section of a valve component unit 440 where said valve functions to enable flow of detergent to reservoir region 487. Components of said valve component unit 440 are labeled 680, 681 and 682 and 621. FIG. 63 shows components 680, FIG. 64 shows components 681 and 682, while FIG. 65 shows spring components 621. Said outlet valve 484 components may preferably configured from Bakelite, melamine or nylon while the spring unit 483 is preferably made from stainless steel for resistance to the humid environment of a faucet. Said inlet valve 440 components may preferably be configured from Bakelite, melamine or nylon while the spring unit 621 is preferably made from stainless steel for resistance to the humid environment of a faucet.

[0221] Referring to FIG. 63, component 680 shows a cylindrically disposed containment 624. Said containment 624 has a region 626 internally spirally grooved for receiving external spirally grooved component 623. Enabled in the interior of said 624 at the region distal to region 626 is a shoulder 629. As shown in FIG. 62 said valve component 682 shown in FIG. 64 is enabled to freely rest on said shoulder 629 of the interior of said 624. Said containment 624 is externally spirally grooved for insertion connection in the conduit component 724 of an internally grooved detergent conduit depicted in FIG. 71. Said region of 624 distal to region 626 flares conically to reservoir region 487 of said cylinder 464.

[0222] Referring to component 681 shown in FIG. 64, what is shown is an annulus shaped component 623 where said component 623 is spirally grooved on its outer curved surface. Said component 623 is concentrically enabled with a circular access opening 622 where said access opening 622 functions as a conduit for detergent fluid. Contiguous and concentric to component 623 is component 625 where said

component 625 is an annulus shaped. The radius of said component 625 is less than the radius of component 623 but greater than the radius of component access conduit opening 622. Said component 625 functions as a guide for spring component 621.

[0223] Referring to component 682 depicted in FIG. 64 said component 682 is disposed to be a composite shape characterized as a solid truncated cone 634 contiguous with a preferably solid cylindrical section 627 where the larger base radius of said solid truncated cone 634 is common to the radius of said cylinder 627. Said preferably solid cylinder 627 is concentrically contiguous with a circular face of an alternate cylinder 631 where said cylinder 631 is enabled with a larger radius than that of said cylinder 627. Said cylinder 631 alternate circular face is further enabled concentrically with an alternate solid cylinder 638 where said alternate solid cylinder 638 is internally spirally grooved for insertion of externally spirally grooved extension 639 of component 632 depicted in FIG. 64. Said externally grooved extension 639 is contiguous with said 632. Said component 632 is disposed to be characterize as a cylinder enabled with a partial spherical component where said cylindrical section is internally spirally grooved. Said cylindrical component 631 is further enabled with conduit channels 628 extending from a circular face of said cylinder 631 to the alternate face of said cylinder 631. Said channels 628 enable the flow of detergent to said region 487 of detergent activation unit 430.

[0224] FIG. 66 shows a relative upward looking view of 681 through a horizontal plane at right angles to the height of cylindrical annulus 625 where said plane contains the line 700 of FIG. 64. FIG. 67 shows a depressive view of 682 through a horizontal plane containing line 702 and where that plane is parallel to a circular face of cylinder 631. FIG. 69 shows a depressive view of 632 with its spirally grooved extension 639 through a plane containing line 704 of FIG. 68 and where that said plane is parallel to a circular face of cylinder 631. FIG. 70 shows a vertical section of the configuration of said component 681 and said component 682 enabled with spring member 621. Extreme ends of said spring 621 enshrouds component 625 and component 627 such that spring 621 is enabled to be compressed when said cylinder 631 is initiated to translate component 632 from said shoulder 629 of 624. Said shoulder functions as a support base for component 631.

[0225] Referring to FIG. 71; what is shown is an embodiment of the detergent component containment unit 450. Said detergent containment unit is enabled with a preferably cylindrical detergent reservoir 716 with means 726 for enabling flow or disabling flow of detergent from said detergent reservoir 716. Component 712 of said detergent reservoir functions as a spirally grooved cap. Removal of said spirally grooved cap 712 enables refilling of said detergent containment reservoir. Said detergent containment reservoir is also enabled with an opening 714. Said opening 714 functions to enable detergent fluid to enter region 487 of 464 when motion of said piston 491 enables a pressure difference between said region 487 and atmospheric pressure in said detergent containment reservoir 716 and said opening 714 enables pressure in said containment reservoir 716 to be that of the atmosphere. Said detergent reservoir 450 is preferably made from polypropylene, polystyrene, polycarbonate plastic or allied compatible transparent or see through recyclable plastic. Transparency or seeing through reservoir better enable management of detergent solution for

timely refilling or replacement of said detergent reservoir 450. Due to anticipated multiple interventions of the reservoir 450 for refilling or replacement a recyclable material for the reservoir 450 is preferred thus enabling environmental, energy and material conservation.

[0226] Referring to FIG. 71, said detergent containment unit is enabled with a preferably cylindrical conduit 722 where said conduit 722 is preferably of a lesser radius than 716 but contiguous with detergent containment reservoir 716. Said conduit is enabled with a thicker wall cylindrical region 720 where the central concentric opening or internal channel 718 is of a lesser radius than the radius of said cylindrical disposed component 722.

[0227] Referring to FIG. 71, the lower region of said 722 is internally spirally grooved for accommodating inclusion of valve component 440 shown in FIG. 62. Referring to FIG. 72 what is shown is a section of an inclusion 732 enabled in said annulus region 720 where said inclusion 732 is configured to accommodate component 726 thus enabling said 726 to initiate enabling or disabling flow of detergent from said containment reservoir 716. Said component 726 may be configured from Bakelite, melamine, nylon or other appropriate compatible material.

[0228] Referring to FIG. 73, what is shown is component 726. Said component 726 is disposed to be characterize as a solid truncated cone 721 contiguous with a rectangular prism 727 where said rectangular prism 727 functions as a lever for initiating motion of said truncated cone 721. Said truncated cone 721 is enabled with a conduit 728 where said conduit is enabled with diametric openings on the curved surface of said truncated cone 721.

[0229] Referring to FIG. 74 what is shown is a top view of an embodiment of the containment reservoir 716. Referring to FIG. 75 what is shown is an orientation of said component 726 in region 732 of 720 when said component 726 disables the flow of detergent from containment reservoir 716. Said conduit 718 is then substantially perpendicular aligned to said conduit 728 such that the flow of detergent fluid is disabled from detergent from containment reservoir 716. Referring to FIG. 76 what is shown is an orientation of said component 726 in region 732 when said component 726 enables the flow of detergent from containment reservoir 716 of detergent containment reservoir 716. Said conduit 718 is then substantially parallel aligned to with conduit 728 such that the flow of detergent fluid is enabled from containment reservoir 716.

[0230] Referring FIG. 77, a clockwise sense of rotation of said lever 20 for the invention will now be enacted. A counterclockwise sense of rotation of lever 20 is a mirrored action for a clockwise sense of rotation. A description for the function of the entity is for an embodiment in which detergent is expelled substantially at about the same time that water is enabled to flow through spout 10 for the washing of hands thus a coordination between the translation of the piston substantially parallel to the axis of cylindrically disposed 464. The compressive length of spring 483, the radius of the crank 460, the travel of valve 482 prior to expulsion of detergent, the retraction of the connecting rod 496 prior to completion of the maximum predetermined angle of rotation of lever 20 are enabled and dimensioned within the maximum predetermined angle of rotation of lever 20 with the radius of the crank 460. An embodiment of the invention in which the leading edge 486 of piston 491 is substantially close to outlet valve 482 in the inactive state is

preferred for timely dispensing of detergent. The washing of the lever **20** is preferably enabled on the counter rotation of said lever **20** and subsequent to the completion of the maximum predetermined angle of rotation of said lever **20** when the crank would have retreated while the lever **20** was being turned from the off position to its maximum predetermined angle of rotation. The vertical line of flow of water from spout **10** is not intercepted by lever **20** when stops **60** and **61** restrict said lever **20** to its predetermined maximum angle of rotation. Thus, as can be seen the utility of the invention disposes the use of the invention to just washing hands with detergent prior to maximum predetermined rotation of said lever **20** when water flows through spout **10** followed by a choice to wash said lever subsequent to completion of the maximum predetermined rotation of lever **20**. Consequently, when moveable disc **102** and stationary disc **112** are configured for maximum opening of **114** for a turn of 90 degrees then a predetermined maximum angle of rotation of lever **20** is greater than 90 degrees thus stops **60** and **61** are not solely disposed to being diametrically opposite for a 90 degree orientation of lever **20** and spout **10**.

[0231] Referring to FIG. **77** what is shown is a sectional view through a vertical plane of an assembled configuration of said crank component unit **410** shown in FIG. **39**, wherein said cartridge component **420** shown in FIG. **40**, said detergent activation unit shown in FIG. **41**, said inlet valve component unit shown in FIG. **42**, and said detergent containment unit **450** shown in FIG. **43** are depicted when said configuration depicted in FIG. **77** exists in an inactive configuration.

[0232] Referring to FIG. **78** what is shown is a section in a horizontal plane of the detergent activation unit **430** in association with the stem **104** and crank unit **410** of FIG. **77**.

[0233] Referring to FIG. **77** the component **486** of piston component **491** is constrained to be in substantial proximity to valve component **482** thus spring component **483** exists in its relaxed configuration, consequently enabling closure of outlet **478** by said valve component **482**. Referring to FIG. **77**, said component **631** rests on shoulders **629** when the reservoir valve **632** is deactivated consequently enabling flow of detergent through conduits **628** to region **487** of **464** when said system is primed for initiation to function where such priming action constitutes a back-and-forth motion of said piston **491** to open and close valve component **632**. Said back and forth motion of piston component **491** initiate opening and closing of valve **632** thus enabling a pressure difference between interior volume **716** of detergent reservoir **450** depicted in FIG. **71** and region **487** of cylinder **454** for the detergent activation unit **430** consequently enabling fluid flow of detergent to **487**.

[0234] Referring to FIG. **78** what is shown is a sectional view in a horizontal plane of an integrated configuration of said crank component unit **410** shown in FIG. **39**, said cartridge component **420** shown in FIG. **40**, said detergent activation unit shown in FIG. **41**, said inlet valve component unit shown in FIG. **42** and said detergent containment unit shown in FIG. **43** when said configuration depicted in FIG. **81** exists in an inactive configuration. Member **495** is positioned within groove **437** of crank unit **410**.

[0235] Referring to FIG. **78a**, what is shown is a diagrammatic representation of the inactive state of the invention. In the inactive state when a lever handle is parallel to component **496** said valve **632** is normally open while said valve **482** is normally closed. Said leading edge **486** of piston

component **491** is substantially close to valve **482** and additionally, outlet spring **483** and inlet spring **621** are in a relaxed state. In FIG. **78a** the leading edge **486** and valve **632** is positioned between the lines **1021** and **1022**. The outlet **478** would be positioned between lines **1022** and **1023**. The outlet **478** diametric dimension is less than that of the separation between lines **1022** and **1023** thus by choosing the travel of the leading edge before opening of valve **478** to be greater than the separation between lines **1021** and **1022** both valves **632** and **482** can be closed to enable build up of pressure in **487** before valve **482** opens to dispel fluid through conduit **478** while valve **632** remains closed.

[0236] It is preferred that detergent be expelled substantially at the same time as the flow of water through spout **10** or immediately prior to the commencement of water flow through spout **10**. Prior expulsion of detergent is enabled when the moveable disc **102** sufficiently blocks area **114** of the stationary disc **112** to enable turning of stem **104** by the lever **20** to enable the action crank **460** and follower **496** to activate detergent expulsion prior to water flow through openings **114** of the stationary disc. Thus, geometric relation between the angular position of the crank **410** on the stem **104**, the lift of said crank **410**, the follower **495**, the leading edge **464** and valve **482** is enabled to allow timely expulsion of detergent within the interval of turning of said lever **20**.

[0237] Referring to FIG. **79** what is shown is a view of an integrated configuration of said crank unit **410** shown in FIG. **39**, said cartridge component **420** shown in FIG. **40**, with a sectional view of said detergent activation unit shown in FIG. **41**, said inlet valve component unit shown in FIG. **42** and said detergent containment unit shown in FIG. **43**. For the configuration of FIG. **79**, when said lever **20** has turned clockwise at a predetermined angle the crank unit **410** is at its maximum displacement and the spring **483** exerts a maximum restoring force on the outlet valve unit **482**. Additionally, said piston **491** is substantially constrained by channel **492** to only motion by translation consequently enabling predictable functioning of the invention.

[0238] Referring to FIG. **79**, the rotation of lever **20** enables beveled edge **488** of said piston **491** to function as a wedge to intercept and initiate vertical translation of valve member **632** thus dislodging inlet valve component **682** shown in FIG. **64** from its seat **629** and thus initiating distortion of spring **621** and thus enabling a restoring force on said spring **621** and consequently enabling member **632** to propagated normal force on said piston unit component **449** shown in FIG. **57**.

[0239] Thus piston unit component **449** shown in FIG. **57** is acted on by an axial force by said outlet spring member **483** and a normal force by inlet spring member **621** thus when said inlet spring member **621** is enabled with a superior spring constant that opposes the influence of spring **483** on said piston unit **449** said piston unit **449** is disposed to be immobilized and consequently disabling any influence of the spring **483** on the rotation stem **104**.

[0240] Additionally, vertical motion of valve member **632** further enables closure of said valve **632** thus disabling flow of liquid detergent through conduits **628** to region **487** of cylinder **464** consequently enabling build up of fluid pressure in region **487** of said cylinder **464** when said lever **20** continues to rotate and said valve member **632** continues to maintain a stress on said piston member **449** when valve **482** is still closed.

[0241] Referring to FIG. 79, said lever 20, said crank unit 460 and said stem 104 substantially function as a rigid body thus clockwise motion of lever 20 enables said crank 410 to translate piston 491 consequently enabling piston extension 486 to intercept and translate and open outlet valve 482 and thus enable ejection of detergent fluid through outlet 478.

[0242] Referring to FIG. 79, said crank 460 on its rise interval would have translated said piston extension a measure substantially that of the length of the crank 460. Thus, spring component 483 would thus have been distorted a displacement substantially that of the length of the crank 460 when said distorted displacement of said spring member 483 is allowable by the physical characteristics of said spring component 483. Said crank member 460 is thus predetermined to execute its maximum translation prior to the execution of the clockwise maximum predetermined angle of rotation of said lever 20.

[0243] At the initial turn of lever 20 when detergent has been enabled and water flows through spout 10 hands are enabled to be washed and the faucet counter turned to its off mode without enabling washing of lever 20.

[0244] Referring to FIG. 80, what is shown is a horizontal cross-section of the configuration depicted in FIG. 79.

[0245] Referring to FIG. 80a, what is shown is the closure of valve 632 and the opening of valve 482 between lines 1022 and 1023. Crank 460 now is at its maximum lift while spring 482 is now exerting a maximum restoring force simultaneously as spring 621 exerts a force on piston component 491. The lever 20 has not yet turned through its maximum predetermined angle. The geometric relation of the stationary disc 112, the moveable disc 102 and the configuration of lever 20 relative to fixed stops 60 and 61 were said fixed stops 60 and 61 are positioned on the upper surface 130 is shown in FIG. 25.

[0246] Referring to FIG. 81, what is depicted is a configuration wherein said lever 20 has executed its clockwise maximum predetermined angle of rotation and said lever 20 is deterred from further rotation by said limiting cartridge body stop. Said connecting rod 496 would have retreated with reduced influence of spring 483 with maximum influence of spring 621 on piston 491 with further rotation restricted by body stop 61 and the force of spring 621 acting on piston 491 thus solvent such as water is enabled to flow without interruption when the invention is being utilized. Valves 632 and 483 are both closed for the configuration of FIG. 81.

[0247] Referring to FIG. 82, what is depicted is a top view configuration of the configuration shown in FIG. 81.

[0248] Referring to FIG. 82a what is shown is a diagrammatic representation in the active state when the lever 20 has been turned through its maximum predetermined angle of turn and the leading edge 464 has retreated from the valve 482 while being acted on by a reduced axial force by spring 483.

[0249] Referring to FIG. 82a, the crank 460 has also now retreated and the choice of the dimension of the cross section of outlet conduit 478 and the separation between line 1022 and line 1023 has enabled the closure of valves 482 and valve 632. The combined force of action by spring 621 on the piston 491 and the reduction of the force of spring 482 coupled with the off set in combination with the action of stop member 61 immobilizes the lever at its maximum

predetermined angle of rotation thus water or any other solvent is enabled to flow the spout 10 without interruption for the washing of hands.

[0250] A counter turn of lever 20 enables said lever 20 to intercept the line of flow 777 from said spout 10 thus enabling washing of the lever and further washing of hands. Further counter turn of said lever to its rest position enables refilling of said region 487 of cylinder 464 in preparation for the utility of the invention.

[0251] Referring to FIG. 82b what is shown is a section in comparison to FIG. 82a that includes outlet 478. When crank 460 retracts at the turning of lever 20 to its maximum predetermined angle outlet 478 is enabled between region 1023 and 1024 thus enabling valve 632 and valve 482 to be in closure mode thus minimizing dripping or leakage.

[0252] When said system depicted in FIG. 81 is being deactivated after use said lever 20 is rotated counter clockwise thus enabling closure of valve 482 while said valve component 632 opens, thus further counter clockwise rotation of lever 20 enables a pressure difference in region 487 and thus enables flow of detergent in said region 487 when said valve member 482 is enabled to be closed prior to the opening of valve 632. Said system is now in a state for its next utility function.

[0253] The detergent activating unit 430 is disposed to be positioned within a circle of influence with the detergent activation unit 430, thus enabling flexibility in the positioning of said detergent activation unit 430 on a faucet mount. Additionally multiple configurations are enabled for timely dispensing of fluid while lever 20 is being rotated.

[0254] Referring to FIG. 83 what is shown is a vertical section in a plane symmetric to the detergent activation unit and the detergent reservoir in relation to the position of the crank unit 410, the cartridge unit and the lever unit in the inactive or closed mode of the invention.

[0255] Referring to FIG. 84, what is shown is a configuration of the lever, the movable disc 102 and the fixed disc for the closed mode for FIG. 83.

[0256] Referring to FIG. 85 what is shown is a vertical section in a plane symmetric to the detergent activation unit and the detergent reservoir in relation to the position of the crank unit 410, the cartridge unit and the lever unit in the active or open mode of the invention.

[0257] Referring to FIG. 86, what is shown is a configuration of the lever, the moveable disc 102 and the fixed disc in the open mode for FIG. 85.

[0258] Referring to FIG. 87 what is shown is a sectional configuration of a substantial cylindrically disposed cartridge mount 820 enabled with slots 821 for mounting the cartridge unit 420. Said member 820 functions to laterally enclose cartridge unit 420 with space between the inner wall of 820 and the outer wall of cartridge body 90 for the cartridge unit 420. Said member 820 also functions to arrest rotary motion of said cartridge body 90 when said cartridge body 90 is associated with said cartridge mount 820 and said tabs 128 of said cartridge body 90 are recessed in symmetrically arranged slots 821 on the perimeter region 819 of said member 820. In an assembled unit of the invention cartridge valve unit 420 is stabilized in region 801 of said cartridge mount 820 when flanges 128 of said cartridge body 130 are recessed in slots 821 of said cartridge mount 820.

[0259] Referring to FIG. 87, said cylindrically disposed member 820 is configured with a grooved upper end 819 with a smaller diameter for accepting bonnet nut 205. Said

member **820** lower end is configured with a larger diameter region **826**. Said larger diameter region **826** is configured with a spirally grooved access **827** for connection to an inlet conduit and a preferably spirally grooved region **831** for connection to an outlet conduit. The outer surface of said cartridge mount **820** between the upper reduced diameter region **819** and the lower larger diameter region **826** is spirally grooved for rigidly attaching said cartridge mount **820** by nuts to a mounting surface. Such a mounting surface **981** is depicted in FIG. **90**.

[0260] Referring to FIG. **88a**, what is shown is interface **861** for cartridge unit **420**. Said interface **861** includes a conduit **70** for allowing water through the openings **114** of the stationary disc **112**. Surface **859** on upper portion of **861** interfaces with the lower surface **123** of fixed disc **112** by way of a seal **88** to provide a seal between member **861** and said stationary disc **112** to prevent leakage between **861** and **820**. A space **900** is enabled between the inner wall of cartridge mount **820** and the outer perimeter of flange surface **858** of said interface **861** consequently enabling output flow from region **110** of the moveable valve **102** to region **829** of said cartridge mount **820** thus enabling fluid flow through **865** to spout **10**; said **861** is then firmly attached to region **827** of cartridge mount **820** when the fixed disc **112** is situated in region **132** of said cartridge body **90**.

[0261] Referring to **88b** what is shown is an interfacial inlet attachment **863** with conduit **70**. Said interfacial conduit has a larger externally spirally grooved diameter region **866** and a smaller externally spirally grooved diameter region **869**. Said outer grooved larger diameter region **866** locks into inner grooved diameter region **824** of **861** and preferably soldered to **861**. Means **868** is provided for insertion of said conduit **863** into grooved region **824** of **861**. The lesser diameter region **869** of **863** interfaces with water line attachments from an inlet source. FIG. **88c** shows a section view of said inlet conduit **863**.

[0262] Referring to FIG. **88d** what is shown is an outlet attachment **865** with conduit **874** for connecting to the spout **10** of the faucet unit. Said outlet conduit **865** is configured with externally spirally grooved end portions **870** and **872** and a region **871** between said grooved region **870** and said grooved region **972**. Said region **871** is enabled with means for insertion into access **831** of cartridge mount **820**. It is preferred that **865** be soldered to cartridge mount **820**.

[0263] Referring to FIG. **88e** when said interface **861** is firmly locked in **827** of the lower region **826** of **820** and said lower surface **123** of fixed disc **112** of cartridge **420** interfaces with the upper surface **859** of **861** and said bonnet nut **205** is attached to **820** to allow the flange **47** of said bonnet nut **205** to be firmly applied over the upper surface **130** of the cartridge body **90** and open region **45** of said bonnet nut **205** fully encloses region **59** of the stem **104** then said stem **104** is disposed to be stabilized for interaction with the disc assembly. A washer **822** interfaces **820** and **205**.

[0264] Referring to FIG. **88e** when the bonnet nut **205** is in firm relation with the upper surface **130** of body **90** of the cartridge unit **420** and said bonnet nut **205** is firmly attached to cartridge mount **820** a seal **88** between the upper surface of **861** and the lower surface **123** of fixed disc **112** is included to prevent leakage between the outer surface of **861** and the inner surface of cartridge **820**. The seal **88** interfaces flange **859** and surface **858** of the upper surface of said **861** with the lower face **123** of the fixed cartridge **112**. The lower

grooved end **864** of said **861** would also be firmly attached at **827** of said cartridge mount **820**.

[0265] Referring to FIG. **88e** said upper region **858** of **861** is configured to accept an elastomeric seal **116** between a ledge **858** of **861** and the lower part **132** of said cartridge body **90**; such a seal **116** functions to prevent leakage between the outer surface of **861** and the interior surface of **820**.

[0266] Referring to FIG. **88e** the attachment between the lower portion **824** of **861** and the lower portion **826** of **820** is enabled with an elastomeric seal **119**. Said seal **119** functions to prevent leakage between **861** and **820**.

[0267] Referring to FIG. **89** what is shown is the embodiment of FIG. **88e** when input mains water supply attachment **863** is attached and preferably soldered to **861** at access opening **824** of said **861** and output water attachment **965** is attached and preferably soldered to **820** at access opening **831** of said cartridge mount **820**. Said conduit **865** functions to direct water to spout **10** when a water line is connected from conduit **865** to spout **10**. Said attachments **865** and **863** are preferably made from copper, brass or stainless steel and soldered to member **820**.

[0268] Referring to FIG. **90** what is shown is a section of embodiment **981** for mounting the invention for use with a wash basin. To secure the detergent activation unit **430** shown in FIG. **48** more ways than one is available; one available means for securing said detergent activation unit involve having access slots **935** for accepting fasteners to which is attached a nut for bonding said detergent activation unit **430** to said embodiment **981**. Said embodiment member **981** has access openings **947** for receiving cartridge housing **820** for enabling nuts **933** to firmly secure said cartridge housing **820** to said mounting base **981**. Said embodiment member **981** has access openings **947** for receiving cartridge housing **820** to allow nuts **933** to firmly secure said cartridge housing **820** to said embodiment **981**.

[0269] Referring to FIG. **90**, said embodiment **981** is enabled with access **939** for receiving spout assembly to allow a nut to secure said spout assembly to said base **981**.

[0270] Referring to FIG. **90**, said mounting base **981** is enabled with opening **943** for setting detergent activation unit **430** and also with access openings **935** for receiving bolts **948**. Said bolts **948** firmly associate undercover plate **930** and detergent activation unit to **430** when a nut **942** is applied to **948** in such a manner that the members **930**, **981** and base member **465** of the cartridge activation unit **430** are contiguous and firmly associated.

[0271] Referring to FIG. **91** what is shown is the embodiment of an under-cover plate **930** for interfacing with mounting base **981** when said **981** is in relation to a wash basin. To secure the under-cover plate **930** more ways than one is available; one available means for securing said detergent activation unit involve having access slots **935** for accepting bolts **948** to which is attached a nut **942** for securing said detergent activation unit **430**. Said embodiment member **930** has access openings **947** where the diameter measurement of said openings **947** is greater than the measurement of the most significant diameter of nuts **933**. Said openings **947** enable access for cartridge mount **820**.

[0272] Referring to FIG. **91** said embodiment **930** is enabled with access **939** for receiving spout assembly **10** and to allow a nut **933** to secure said spout assembly **10** to said base **981** and to said under-cover plate **930**.

[0273] Referring to FIG. 91, under-cover plate 930 is enabled with opening 943" for setting detergent activation unit 430 and also with access openings 935" for receiving bolts 948. Said bolts 948 firmly associate undercover plate 930 and detergent activation unit to 930 when a nut 942 is applied to 948 in such a manner that the members 930, 981 and base member 465 of the cartridge activation unit 430 are contiguous and firmly associated.

[0274] FIG. 92 what is shown is an escutcheon embodiment 940 for mounting on surface 981 when said embodiment is in relation to a wash basin. To secure the escutcheon 940 more ways than one are available; one available means for securing said escutcheon 940 involve having access slots 955 for accepting lever assembly 304 configured so that when said lever assembly is attached to region 58 of stem portion 57 of stem 104 the lower surface of said lever assembly abuts the surface of escutcheon embodiment 940. To further secure the escutcheon 940, or protective covering 940, when spout 10 is included in opening 953 and escutcheon 975 is attached to the lower end of said spout 10 the lower edge of escutcheon 975 abuts the surface of escutcheon 940.

[0275] Said escutcheon 940 shown in FIG. 92 is also enabled with access opening 957 were said access opening 957 accepts lower portion 724 of detergent reservoir unit 450 when said lower portion 724 of the said detergent reservoir unit 450 is attached to region 487 of the detergent activation unit 430, a nut for securing said detergent activation unit 430. Referring to FIG. 92 access opening 951 enables availability of detergent when outlet conduit 478 is accepted by said opening 951 of said detergent activation unit 430.

[0276] Referring to FIG. 92a what is shown is a spout assembly 975 for inclusion with spout 10 where said spout escutcheon 977 of spout assembly 975 is enabled to abut an elastomeric washer 976 and said elastomeric washer 976 abuts surface 981 in an assembly of the invention and is in a relation to the lower region of said spout 10. Nut 944 is applied to the under surface of 981 to firmly attached spout assembly 975 to said 981. An elastomeric gasket 945 is preferred between said nut 944 and the under surface of 981 in the assembly of the spout assembly 975. Opening 939" of undercover gasket 930 is enabled to accommodate the diametric dimensions of nut 944.

[0277] Referring to FIG. 93 what is shown is an explosion of nut 933, rubber washer 934 and cartridge housing 820, the detergent activation unit 430, the fasteners 948 and nuts 94 in relation to mounting base 981.

[0278] FIG. 93a shows a front view of said nut 933, said washer 934 and said cartridge housing 920 assembled on faucet mounting base 981 shown in FIG. 93.

[0279] Referring to FIG. 94 what is shown is the spatial positioning of order of assembly of detergent activation unit 430, crank unit 410 and cartridge unit 430, bonnet nut 205 and stop attachment 404 for an embodiment in which the detergent activation unit 430 is activated when a right hand-side lever 20 is rotated. The lever handle 305 is not shown.

[0280] Referring to FIG. 94a shows the assembly of FIG. 94 with escutcheon 940.

[0281] Referring to FIG. 94b shows the assembly of FIG. 94a with spout assembly 975 and lever handle 304. Screws for attaching lever unit 304 and escutcheon for said screws are not shown.

[0282] Referring to FIG. 95, what is shown is the configuration of FIG. 94b assembled with spout unit 10, lift rod 18 and bonnet nut 205 were said bonnet nut 205 is firmly associated with cartridge mount 820. When said bonnet nut 205 is firmly associated with cartridge mount 820 and region 45 of said bonnet nut 205 surrounds region 59 of stem assembly 104 optimum stability of said stem 104 is facilitated. The outer surface of region 59 of stem unit 104 is enabled to rotate substantially frictionally against the inner surface 45 of the bonnet nut by the application of a Teflon coating to said surface 59 of stem unit 104.

[0283] Referring to FIG. 96, what is shown is the configuration of FIG. 95 assembled with the detergent reservoir unit 450 in region 487. Said access opening 487 is spirally grooved to be compatible with the externally spirally grooved portion 724 of reservoir unit 450 such that when portion 724 is spiraled into access 487 said detergent reservoir 450 is firmly attached to the detergent activation unit 430.

[0284] Referring to FIG. 97, what is shown is an exploded conceptual view of an embodiment of the invention with under cover plate 930. The diameter of openings 947 of said undercover plate 930 is superior to that of the most significant diameter of nut 933 thus minimizing interaction influence with the cartridge unit when the detergent activation unit is being activated.

[0285] Additionally, said under cover plate 930 interfaces securely with mounting 981 when nut 944 is attached to region 955 of spout 10 shown in FIG. 9; in such a manner said nut 944 functions to secure 930 to the lower face 981. Said under-cover plate is further stabilized when screws 948 are introduced from the upper surface of 498 through openings 497 of said 498 and then through openings 935 of 981 and then through openings 935" of 930 thus enabling screw fasteners to be chosen and be applied to firmly bond together the body 465 the of detergent activation unit 430, the mounting block 981 and said undercover plate 930 by one or more screw nut fasteners 942.

[0286] As such and in summary, the figures depict a washable faucet lever with crank-activated detergent dispenser having a lever cartridge 420 having a stem 104 operably coupled to a hand lever 20 configured to rotate in a lever axis of rotation and operably coupled to a two-bar linkage configured to rotate in a linkage axis of rotation parallel to and simultaneous with the lever axis of rotation. Said another way, the piston 491 is configured to translate in a single and linear axis of translation within the enclosed region 468 upon rotating the hand lever 20 in the lever axis of rotation to selectively and simultaneously displace the reservoir valve 632 and outlet valve 482. Stated differently, the piston 491 is configured to translate within the enclosed region 468 upon rotating the hand lever 20 in the lever axis of rotation and in either a counterclockwise or clockwise rotation to selectively and simultaneously displace the reservoir valve 632 and outlet valve 482, i.e., one impetus or rotation of the hand lever 20 will cause the valves 632, 482 to open (or close).

[0287] The washable faucet lever with crank-activated detergent dispenser also includes a detergent reservoir member 450 configured to house a liquid detergent therein, having an internal channel 718 configured to receive the liquid detergent, and having a reservoir valve 632 coupled to the detergent reservoir member 450 and operably configured to selectively translate to open and permit flow of the liquid

detergent through the internal channel 718 and to close and restrict flow of the liquid detergent through the internal channel 718. Further, the dispenser also includes an encasement member 498 defining an enclosed region 468 therein, defining an outlet 478, having an outlet valve 482 disposed within the enclosed region 468 and operably configured to selectively translate to open and permit flow of the liquid detergent through the outlet 478, to close and restrict flow of the liquid detergent through the outlet 478, and biased with a spring 483 retained by the encasement member 498.

[0288] Further, the washable faucet lever with crank-activated detergent dispenser may also include a piston 491 disposed within the enclosed region 468, coupled to the two-bar linkage, and operably configured to translate within the enclosed region 468 upon rotating the hand lever 20 in the lever axis of rotation to selectively and simultaneously displace the reservoir valve 632 and outlet valve 482 to permit flow of the liquid detergent through the internal channel 718, into the enclosed region 468, and out through the outlet 478. In one embodiment, the linkage axis of rotation is coaxial with respect to the lever axis of rotation.

[0289] In one embodiment of the present invention a mounting base 981 is utilized and that defines an opening with the lever cartridge 420 disposed therein and coupled thereto with at least one fastener, coupled to the encasement member 498 with at least one fastener, and defines another opening with a faucet member disposed therein. The faucet member includes a spout 10 with an opening configured to emit liquid therethrough and is coupled (i.e., directly or indirectly) to the mounting base 981 with at least one fastener. The dispenser will also include a protective covering 940 defining an opening with the lever cartridge 420 disposed therein, covering the encasement member 498, defining an opening with the faucet member disposed therein, and defining an opening with the detergent reservoir member 450 disposed therein.

[0290] In some embodiments, the protective covering 940 defines an internal channel fluidly coupled to the outlet 478 and configured to permit flow of the liquid detergent therethrough and the reservoir valve 632 is biasedly coupled to the detergent reservoir member 450 with a spring. Also in some embodiments, the internal channel 718 is spatially configured in the detergent reservoir member 450 to enable gravity-based receipt of the liquid detergent therein.

[0291] Referring to FIGS. 98a, 98b and 98c, what is shown is another embodiment of the water line system into the cartridge unit and water line system out of the cartridge unit for the invention. The embodiment shown for FIG. 98a, FIG. 98b and FIG. 98c are an alternative for what has been previously described and thus only information related to this embodiment for installation and assembly needs to be described. The valve chamber is composed of an inlet tube 961', a main tube body 920' and an outlet/inlet sleeve 963'. The inlet tube 961' is preferably made from copper or brass or stainless steel and formed with an inlet tunnel 70'. The outer circumference of the upper end of the inlet tube 961' is formed with an annular engaging section 958' with a larger diameter. The middle portion is milled to form a stepped section. An annular flow guiding rib 967' with a larger diameter is adjacent to the lower side of the stepped section. A water scaling ring 119' is fitted under the flow guiding rib 967'. The lower end of the inlet tube 961' under the flow guiding rib 967' is milled with a locking outer thread section 964'. The main tube body 920' is also made from copper,

brass or stainless steel and milled to form an internal receiving tunnel 901'. The inner circumference of the receiving tunnel 901' near the lower end thereof is formed with an annular stop section 925'. The outer circumference of the upper end of the main tube body 920' is formed with an outer thread section 921' with a smaller diameter. The edge of the upper opening is cut with two diametric notches 921'. The middle portion of the main tube body 920' is formed with a main outer thread section 903' with a certain length. The lower end of the main tube body 920' under the main thread section 903' is a smaller diameter section milled with fastening outer thread section 902'. A water sealing ring 117' is fitted on the small diameter section. The out/let sleeve 963' is made by pressing cast. The center thereof is with a stepped adapting flow passage 910. The opening of the larger diameter section of the adapting flow passage 910' is formed with a fastening inner thread section 904' and the smaller diameter section thereof is formed with a locking inner thread section 909'. A lower side of the outlet/inlet tube 50' is formed with an inlet tube 908' and a lateral side thereof is formed with an outlet tube 906'. The inlet tube 908' and the outlet tube 906' both communicate with the adapting flow passage 910'. In addition, the outer circumference of the end of the inlet tube 908' is formed with an inlet outer thread section 907' while the outer circumference of the end of the tube 906' is formed with an outlet outer thread section 905'.

[0292] Referring to FIG. 99, when assembled the inlet tube 961' is first fitted with the receiving tunnel 901' of the main tube body 920' with the engaging section 958' engaged with stop section 925' of main tube body 920'. At this time, the inlet tube 961' is preliminary assembled with the main tube body 920' with locking outer thread section 964' of the inlet tube 961' extending out of the main tube body 920' to be screwed in with the locking inner thread section 909' of the outlet/inlet sleeve 963'. The locking outer thread section 47' 902' of the main tube body 920' is screwed with the fastening inner thread section 904' of the outlet/inlet tube 963'. Under such circumstances the inlet tube 961', main tube body 920' and the outlet/inlet tube 963' are integrally associated together. A valve chamber 801' with a certain depth is defined between the upper end of the engaging section 958' of the inlet tube 961' and the upper opening of the receiving tunnel 901' of the main tube body 920'. Also, an annular outlet passage 900' with a certain size is defined between the outer circumference wall of the tube 961' and the inner circumference walls of the main tube body 920' and the large diameter section of the outlet/inlet sleeve 963'. At this moment, the insertion of the cartridge unit and stabilization of the cartridge by bonnet nut completes the assembly for the functioning of the cartridge chamber.

[0293] Referring to FIG. 100, what is shown is an alternative embodiment of the stem portion 57' for the cartridge 604. Said stem portion 57' has an upper region 258 configured as a rectangular prism for inserting said portion 258 into rectangular access 234 of the lever assembly 814; said lever assembly 814 is shown in FIG. 103. Contiguous with portion 258 of stem portion 258 is cam attachment region 252; said region 252 fits into rectangular access opening 234 of cam unit 810. Crank unit 810 is shown in FIG. 106. Contiguous with region 252 of stem portion 57' is a cylindrical region 259 were said cylindrical region 259 fits into smooth inner circular opening 45 of bonnet nut 205 shown in FIG. 105. Said bonnet nut 205 when assembled is firmly attached to cartridge housing 920 and thus said bonnet nut

is disabled from motion when the lever assembly **814** is rotated. A section view of bonnet nut **205** is shown in FIG. **104**. Referring to FIG. **101**, what is shown is an alternative embodiment **89** for intercepting body stops **260** and **261** and thus restricting the rotation of lever unit to its maximum predetermined angle of rotation. Said embodiment **89** has rectangular opening that intimately fits over region **256** of region **57'** of the stem unit **104'** shown in FIG. **100**. FIG. **102** shows a configuration of **89** fitted onto the upper surface **230** of the cartridge body **90**.

[0294] Referring to FIG. **107**, when the cartridge stem is already inserted in body passage **68** member **89** is first attached to region **256**; bonnet nut **814** is then applied to the cartridge housing **920** to stabilize the cartridge unit; bonnet nut **814** now completely encloses member **89**. Subsequently cam body unit **810** is attached above the bonnet nut and followed by attachment of lever assembly **814**.

[0295] Other features and advantages will become apparent from the following description preferably taken relative to the accompanying drawings, which in combination discloses an embodiment of the invention. For example, FIG. **108** shows an alternative embodiment for interacting with the detergent activation unit.

What is claimed is:

1. A washable faucet lever with crank-activated detergent dispenser comprising:

- a lever cartridge having a stem operably coupled to a hand lever configured to rotate in a lever axis of rotation and operably coupled to a two-bar linkage configured to rotate in a linkage axis of rotation parallel to and simultaneous with the lever axis of rotation;
- a detergent reservoir member configured to house a liquid detergent therein, having an internal channel configured to receive the liquid detergent, and having a reservoir valve coupled to the detergent reservoir member and operably configured to selectively translate to open and permit flow of the liquid detergent through the internal channel and to close and restrict flow of the liquid detergent through the internal channel;
- an encasement member defining an enclosed region therein, defining an outlet, having an outlet valve disposed within the enclosed region and operably configured to selectively translate to open and permit flow of the liquid detergent through the outlet, to close and restrict flow of the liquid detergent through the outlet, and biased with a spring retained by the encasement member; and
- a piston disposed within the enclosed region, coupled to the two-bar linkage, and operably configured to translate within the enclosed region upon rotating the hand lever in the lever axis of rotation to selectively and

simultaneously displace the reservoir valve and outlet valve to permit flow of the liquid detergent through the internal channel, into the enclosed region, and out through the outlet.

2. The washable faucet lever with crank-activated detergent dispenser according to claim **1**, wherein the piston is configured to translate in a single and linear axis of translation within the enclosed region upon rotating the hand lever in the lever axis of rotation to selectively and simultaneously displace the reservoir valve and the outlet valve.

3. The washable faucet lever with crank-activated detergent dispenser according to claim **1**, wherein the piston is configured to translate within the enclosed region upon rotating the hand lever in the lever axis of rotation and in one of a counterclockwise or clockwise rotation to selectively and simultaneously displace the reservoir valve and the outlet valve.

4. The washable faucet lever with crank-activated detergent dispenser according to claim **1**, wherein the linkage axis of rotation is coaxial with respect to the lever axis of rotation.

5. The washable faucet lever with crank-activated detergent dispenser according to claim **1**, further comprising:

- a mounting base defining an opening with the lever cartridge disposed therein and coupled thereto with at least one fastener, coupled to the encasement member with at least one fastener, and defining an opening with a faucet member disposed therein, having a spout with an opening configured to emit liquid therethrough, and coupled thereto with at least one fastener; and
- a protective covering defining an opening with the lever cartridge disposed therein, covering the encasement member, defining an opening with the faucet member disposed therein, and defining an opening with the detergent reservoir member disposed therein.

6. The washable faucet lever with crank-activated detergent dispenser according to claim **5**, wherein the protective covering defines an internal channel fluidly coupled to the outlet and configured to permit flow of the liquid detergent therethrough.

7. The washable faucet lever with crank-activated detergent dispenser according to claim **1**, wherein the reservoir valve is biasedly coupled to the detergent reservoir member with a spring.

8. The washable faucet lever with crank-activated detergent dispenser according to claim **1**, wherein the internal channel is spatially configured in the detergent reservoir member to enable gravity-based receipt of the liquid detergent therein.

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